

Dynamically suppressing cavity dephasing induced by frequency fluctuations of a nonlinear mode

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High-coherence superconducting cavities are a promising platform for quantum information, offering excellent coherence and negligible intrinsic dephasing noise. However, cavity control relies on nonlinear Josephson elements, frequency fluctuations of which induce cavity dephasing, potentially limiting control fidelities and breaking the noise bias used for error-correction protocols. In this work, we propose a hardware-efficient protocol to protect the cavity from such inherited dephasing, using only a weak off-resonant microwave drive on the nonlinear element. As a concrete setting, we analyze a 3D superconducting cavity dispersively coupled to a flux-tunable transmon (FTT) subject to $1/f$ flux noise. Monte Carlo simulations with realistic parameters show that a weak drive can extend the cavity dephasing time by more than an order of magnitude without incurring detrimental effects from the additional drive.

I. INTRODUCTION

Superconducting microwave cavities are a leading platform for quantum information processing because they enable information to be encoded in harmonic-oscillator modes with exceptionally long coherence times [1–5]. Their linear character suppresses many parasitic nonlinear loss channels, so photon damping is the dominant error mechanism. At the same time, the large oscillator Hilbert space enables hardware-efficient bosonic encodings. These features make cavities attractive as quantum memories and logical qubits, with bosonic quantum error-correction (QEC) schemes to protect information against dominant physical errors [6–19]. Many protocols rely not only on long coherence, but also on a strong bias in the noise: photon loss is far more likely than dephasing. This biased error landscape is central to cavity encodings such as dual-rail qubits, whose fault-tolerance advantages derive from converting dominant single-photon loss into a detectable erasure channel [20]. Preserving the photon-loss-dominated bias is therefore as important as extending coherence itself.

Universal control of cavity modes generally requires dispersive coupling to a nonlinear element [21–25]. A family of flux-tunable circuits such as the Superconducting Nonlinear Asymmetric Inductive eLement (SNAIL) [24], the Flux-Tunable Transmon (FTT) [26], and the Linear INductive Coupler (LINC) [25] enable beam-splitter interactions through three-wave mixing, providing a practical route to high-fidelity bosonic gates. Activating these interactions often requires biasing the nonlinear element away from its sweet spot, where flux noise causes fluctuations of the nonlinear-element frequency. In the dispersive regime, these fluctuations produce Lamb-shift fluctuations, leading to cavity-frequency

fluctuations and thus cavity dephasing. This inherited dephasing is especially harmful because it introduces an error channel that directly competes with the favorable photon-loss-dominated bias. As the coupling strength between the cavity and the nonlinear element is increased to speed up gates, this inherited dephasing can become a leading decoherence mechanism, degrading the bias structure that bosonic QEC and dual-rail encodings rely on for their advantage.

A commonly used strategy is to idle the nonlinear element at a flux sweet spot and tune it away from the sweet spot only when operations are needed. In three-dimensional architectures, fast flux tuning can be implemented using magnetic hoses [27, 28] or on-chip flux lines [29, 30]. However, these schemes increase hardware complexity and can slow the operation of interest. Moreover, tuning flux might risk leakage. Thus, although sweet-spot biasing suppresses flux sensitivity, fast on-demand bosonic control still relies on flux excursions away from the sweet spot.

This work introduces an alternative method to suppress inherited cavity dephasing without changing the dc flux bias. An off-resonant drive applied to the nonlinear mode induces a photon-number-dependent AC-Stark shift of the cavity transition frequency. With suitable drive parameters, this shift renders the cavity frequency first-order insensitive to flux noise, thereby creating a driven sweet spot. This mechanism differs from the dynamical sweet spots studied for qubits in [31, 32], where a resonant or near-resonant drive on the qubit itself engineers the sweet spot. Here, by contrast, the drive acts on the nonlinear mode rather than on the protected cavity mode, and it introduces only a small correction to the cavity decay rate, while substantially suppressing inherited dephasing. For a representative parameter set, Monte Carlo simulations show a $\sim 20\times$ improvement in the cavity dephasing time, from $\sim 100\ \mu\text{s}$ to $\sim 2\ \text{ms}$ with negligible effect on decay time. Our scheme is also different from dynamical decoupling [33]. The scheme does not require cavity gate sequences and therefore avoids

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the $\sim 1 \mu\text{s}$ overhead of typical cavity gates [34, 35].

The paper is organized as follows. Section II introduces the drive-induced suppression mechanism and derives the condition for first-order insensitivity to flux fluctuations. Section III develops a theory of the total cavity dephasing rate, including relaxation and $1/f$ flux noise, and validates the analytical predictions numerically. Section IV applies the framework to bosonic encodings, emphasizing implications for binomial and dual-rail architectures. Section V summarizes the conclusions, with technical details deferred to the Appendices. While the analysis focuses on the FTT–cavity system as a concrete example, the underlying mechanism extends broadly to flux-tunable nonlinear elements.

II. DEPHASING MITIGATION BY DRIVE

While the dephasing-suppression scheme presented here applies broadly across superconducting circuit architectures, we focus on a flux-tunable transmon (FTT) dispersively coupled to a cavity to make explicit the mechanism by which flux noise induces cavity dephasing.

A. System setup

The Hamiltonian of the FTT–cavity system is

$$H(t) = H_{\text{static}} + H_{\text{noise}}, \quad (1)$$

$$H_{\text{static}} = \omega_b b^\dagger b + \frac{K}{2} b^{\dagger 2} b^2 + \omega_c c^\dagger c + g(b + b^\dagger)(c + c^\dagger), \quad (2)$$

$$H_{\text{noise}} = \delta\omega_b(t) b^\dagger b. \quad (3)$$

Here, b (c) annihilates an excitation in the FTT (cavity) mode; K is the FTT anharmonicity; and g is the coupling strength. Flux noise can cause both depolarization and dephasing. Considering flux noise as small-amplitude $1/f$ noise, we neglect depolarization at GHz-frequency transitions triggered (see Appendix A for more details). Thus, the noise dominantly causes an FTT-frequency fluctuation $\delta\omega_b(t) = \delta\Phi(t) \partial\omega_b/\partial\Phi$, where Φ is the static flux bias and $\delta\Phi(t)$ is its fluctuation.

In the dispersive frame [36], the effective Hamiltonian for H_{static} takes the form

$$H_{\text{disp}} = \bar{\omega}_b b^\dagger b + \frac{K}{2} b^{\dagger 2} b^2 + \bar{\omega}_c c^\dagger c + \chi b^\dagger b c^\dagger c, \quad (4)$$

where χ is the dispersive shift and $\bar{\omega}_b$ and $\bar{\omega}_c$ are the Lamb-shifted FTT and cavity frequencies, which depend on the detuning $\Delta_{bc} = \omega_b - \omega_c$:

$$\chi \approx 2 \frac{g^2}{\Delta_{bc}^2} K, \quad \bar{\omega}_b \approx \omega_b + \frac{g^2}{\Delta_{bc}}, \quad \bar{\omega}_c \approx \omega_c - \frac{g^2}{\Delta_{bc}}. \quad (5)$$

The noise Hamiltonian H_{noise} primarily induces dephasing between eigenstates of H_{disp} (see Appendix H 2). In

particular, because Δ_{bc} depends on $\bar{\omega}_c$, FTT-frequency fluctuations $\delta\omega_b(t)$ lead to cavity-frequency fluctuations

$$\delta\bar{\omega}_c(t) = \frac{\partial\bar{\omega}_c}{\partial\omega_b} \delta\omega_b(t) \approx \frac{g^2}{\Delta_{bc}^2} \delta\omega_b(t), \quad (6)$$

thereby producing cavity dephasing. This is summarized in Fig. 1(b).

B. Driven sweet spots

FTT frequency fluctuations $\delta\omega_b(t)$ are converted into cavity-frequency fluctuations $\delta\bar{\omega}_c(t)$ through the Lamb shift, producing cavity dephasing. This dephasing channel can be suppressed by driving the FTT [Fig. 1(a)], which induces a photon-number-dependent energy shift that can be tuned to cancel the Lamb-shift fluctuations.

In the frame rotating at the drive frequency ω_d , the driven dispersive Hamiltonian reads

$$H_R = \Delta_{bd} b^\dagger b + \frac{K}{2} b^{\dagger 2} b^2 + \bar{\omega}_c c^\dagger c + \chi b^\dagger b c^\dagger c + \Omega_0 (b + b^\dagger), \quad (7)$$

where $\Delta_{bd} = \bar{\omega}_b - \omega_d$ is the FTT–drive detuning. The drive couples $|g, m\rangle$ and $|e, m\rangle$ at fixed cavity photon number m , producing level repulsion that lowers the energy of $|g, m\rangle$. The dispersive interaction offsets the detuning by $m\chi$, so the relevant detuning is $\Delta_m \equiv \Delta_{bd} + m\chi$ and the drive-induced energy shift becomes photon-number dependent. Examples for $m = 0, 1$ are shown in Fig. 1(c), and this m -dependent shift can be tuned to cancel fluctuations of the static Lamb shift [Fig. 1(d)].

An analytic expression follows in the weak-drive regime, assuming (i) $\Omega_0 \ll \min_m |\Delta_m|$ over the photon numbers of interest, so that the drive does not strongly hybridize $|g, m\rangle$ and $|e, m\rangle$, and (ii) $|\Omega_0| \ll |K + 2\Delta_{bd}|$, so that hybridization with the state $|f, m\rangle$, remains weak. Under these conditions, a Schrieffer–Wolff transformation (Appendix B) diagonalizes H_R . Retaining terms up to $\mathcal{O}(\Omega_0^2/\Delta_m^2)$ and projecting onto the $\{|g, m\rangle, |e, m\rangle\}$ subspace gives

$$H_d = \tilde{\Delta}_{bd} \sigma^+ \sigma^- + \bar{\omega}_c c^\dagger c - \sum_m \frac{\Omega_0^2}{\Delta_m} |m\rangle\langle m| + \sigma^+ \sigma^- \left[\chi c^\dagger c + \sum_m \frac{2\Omega_0^2}{\Delta_m} |m\rangle\langle m| \right], \quad (8)$$

where $\sigma^+ \sigma^- \equiv |e\rangle\langle e|$ and $|m\rangle$ denotes a cavity Fock state. The drive renormalizes the FTT detuning, $\tilde{\Delta}_{bd} \equiv \Delta_{bd} - \Omega_0^2/K$, and generates an m -dependent shift of the cavity transition frequency together with an additional Stark shift. The essential point is that this drive-induced shift can be tuned to cancel the flux-sensitive static Lamb shift, thereby creating a driven sweet spot.

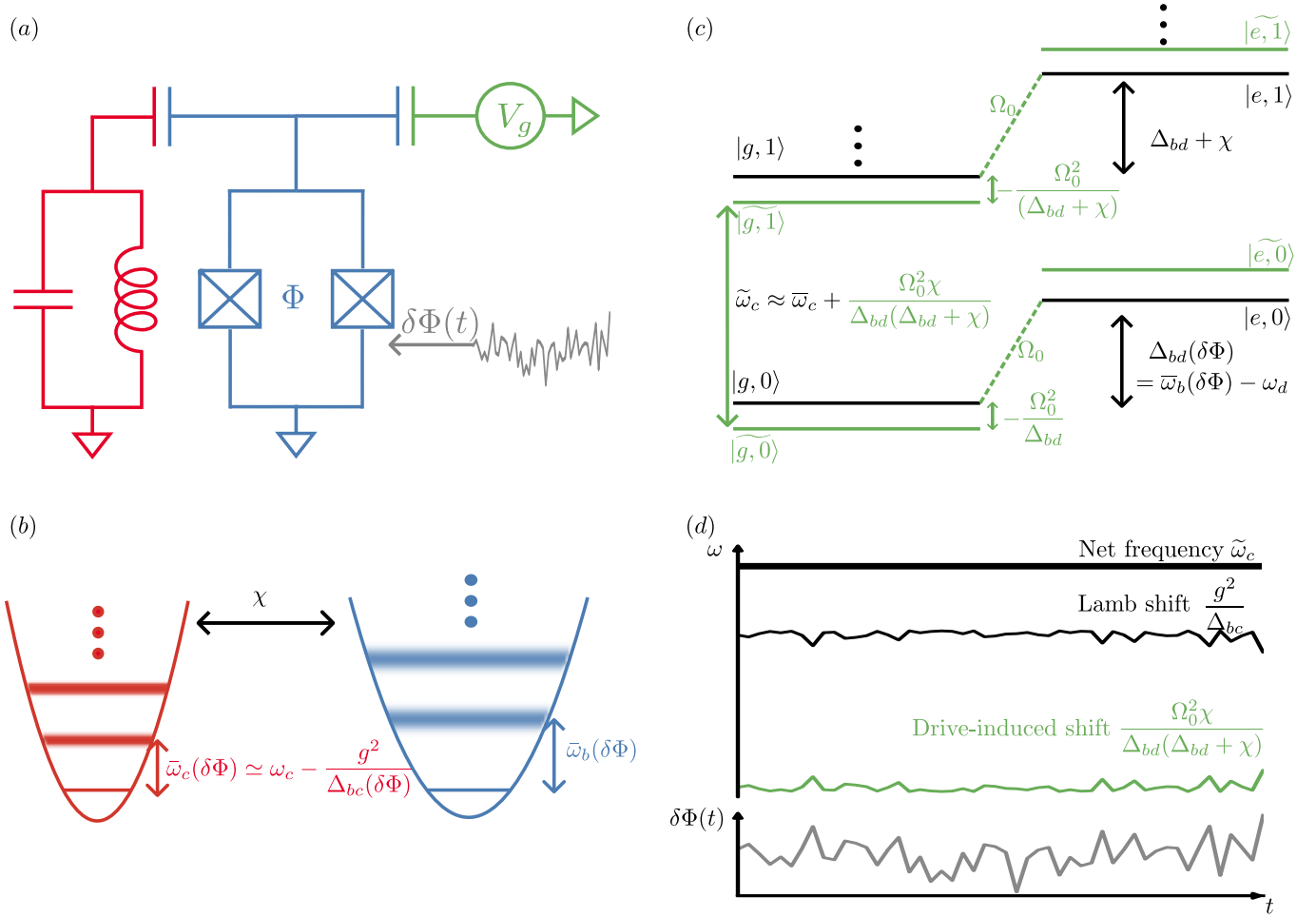


Figure 1. **Scheme for engineering a cavity frequency at a driven sweet spot.** (a) Circuit schematic of a flux-tunable transmon (FTT) dispersively coupled to a superconducting cavity and driven by a coherent voltage source V_g . Flux noise $\delta\Phi(t)$ modulates the FTT frequency and provides the dominant noise channel. (b) Noise transduction via the Lamb shift. The bare modes have frequencies ω_c and $\omega_b(\delta\Phi)$ with flux-dependent detuning $\Delta_{bc}(\delta\Phi) = \omega_b(\delta\Phi) - \omega_c$. The coupling g hybridizes the modes, yielding dressed frequencies $\tilde{\omega}_{b/c}(\delta\Phi)$; the cavity frequency acquires a Lamb shift $-g^2/\Delta_{bc}$ and therefore inherits sensitivity to $\delta\Phi(t)$. (c) Drive-induced, photon-number-dependent shift in the frame rotating at ω_d . A coherent drive of amplitude Ω_0 hybridizes the manifolds $|g, m\rangle$ and $|e, m\rangle$, producing level repulsion (green) that renormalizes the cavity transition frequency in an m -dependent manner. (d) First-order noise cancellation at the driven sweet spot. In the presence of $\delta\Phi(t)$ (gray), the static Lamb-shift contribution to the cavity frequency (black) is compensated by the drive-induced shift (green) at an appropriate drive, yielding a net cavity frequency $\tilde{\omega}_c$ (thick black) that is first-order insensitive to flux fluctuations (a flux-insensitive point in the sense of vanishing first-order susceptibility).

To quantify the resulting cavity-frequency fluctuation, we define the net cavity transition frequency between adjacent cavity levels as

$$\tilde{\omega}_{c,m \rightarrow m+1} = \bar{\omega}_c - \Omega_0^2/\Delta_{m+1} + \Omega_0^2/\Delta_m. \quad (9)$$

We use a bar for Lamb-shifted frequencies and a tilde for drive-renormalized transition frequencies. FTT frequency fluctuation $\delta\omega_b(t)$ then induces

$$\delta\tilde{\omega}_{c,m \rightarrow m+1}(t) = D_{m \rightarrow m+1} \delta\omega_b(t) \quad (10)$$

where $D_{m \rightarrow m+1} \equiv \frac{\partial \tilde{\omega}_{c,m \rightarrow m+1}}{\partial \omega_b}$ is the susceptibility of the net cavity frequency to FTT-frequency fluctuation. It

has two contributions: a bare term inherited from the Lamb-shift dependence $\bar{\omega}_c(\omega_b)$, and a drive-induced term from the dressing shift through $\Delta_m = \Delta_{bd} + m\chi$. A sweet spot occurs when these two terms cancel. Specifically, the physics of the cancellation differs in two regimes.

a. Unselective Regime. In this regime, the drive detuning $\Delta_{m^*} \equiv \Delta_{bd} + m^*\chi$ satisfies $|\Delta_{m^*}| \gg |\chi|$, so the drive couples $|g, m^*\rangle \leftrightarrow |e, m^*\rangle$ and neighboring photon-number pairs with comparable strength. As a result, the effective cavity nonlinearity near m^* is suppressed. For $m^* = 0$, expanding the denominators $\Delta_m^{-1} = (\Delta_{bd} + m\chi)^{-1}$ to leading order in $|\chi/\Delta_{bd}|$ yields

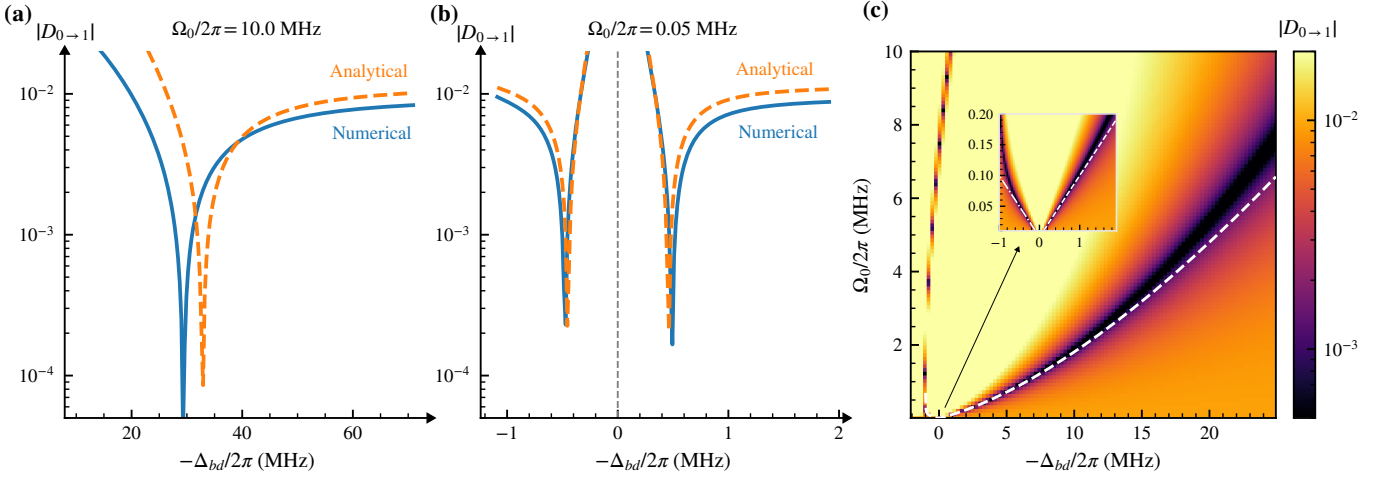


Figure 2. Drive-induced flux sweet spots, defined by $D_{0\rightarrow 1} = 0$. (a) Unselective regime at $\Omega_0/2\pi = 10$ MHz, showing a single sweet spot. (b) Selective regime at $\Omega_0/2\pi = 0.05$ MHz, showing two symmetric sweet spots about the effective resonance. (c) Two-dimensional map of $D_{0\rightarrow 1}$ versus drive amplitude Ω_0 and detuning Δ_{bd} ; Dashed curves denote analytical sweet-spot conditions from Eqs. (9) and (16) for the main panel and the inset, respectively.

the approximate effective Hamiltonian

$$H_d \approx \tilde{\Delta}_{bd} \sigma^+ \sigma^- + \tilde{\omega}_{c,0\rightarrow 1} c^\dagger c + \chi \left(1 - \frac{2\Omega_0^2}{\Delta_{bd}^2} \right) \sigma^+ \sigma^- c^\dagger c, \quad (11)$$

up to constant energy offsets, where

$$\tilde{\omega}_{c,0\rightarrow 1} = \bar{\omega}_c + \frac{\Omega_0^2 \chi}{\Delta_{bd}^2}. \quad (12)$$

The corresponding susceptibility of the net cavity frequency is

$$D_{0\rightarrow 1} \approx \frac{g^2}{\Delta_{bc}^2} \left[1 - \frac{4\Omega_0^2 K}{\Delta_{bd}^3} \right]. \quad (13)$$

The far-detuned sweet spot is therefore set by $D_{0\rightarrow 1} = 0$, which gives the single solution

$$\Delta_{bd} = \sqrt[3]{4\Omega_0^2 K}. \quad (14)$$

This prediction is confirmed by Fig. 2(b). Because the rotating-frame spectrum of H_R approximates the Floquet quasienergies (defined modulo ω_d ; see App. C), the analytical susceptibility is compared against a Floquet calculation using the full-model Hamiltonian (App. D). For $\Omega_0/2\pi = 10$ MHz, $D_{0\rightarrow 1}$ approaches the bare Lamb-shift value $g^2/\Delta_{bc}^2 \approx 0.01$ when the drive is far detuned from the FTT frequency $\bar{\omega}_b$. As ω_d is tuned toward $\bar{\omega}_b$, a single sweet spot appears on the $\omega_d > \bar{\omega}_b$ side, consistent with the negative anharmonicity of the FTT. The characteristic detuning is consistent with Eq. (14), which predicts $\Delta_{bd}/2\pi \sim -34$ MHz for $K/2\pi \approx -0.1$ GHz used here. The residual offset between the analytical and numerical dip positions at this drive strength arises because higher-order corrections to the Hamiltonian Eq. (8) become non-negligible. Closer to resonance, $D_{0\rightarrow 1}$ rises again as the drive-induced energy shift dominates the susceptibility.

b. Selective Regime. In this regime, the drive is closest to resonance for a particular photon number m^* , so that $|\Delta_{m^*}| \ll \chi$. Consequently, only the hybridization between $|g, m^*\rangle$ and $|e, m^*\rangle$ contributes appreciably to the cavity frequency, giving

$$\tilde{\omega}_{c, m^* \rightarrow m^*+1} \approx \bar{\omega}_c + \frac{\Omega_0^2}{\Delta_{m^*}}. \quad (15)$$

Differentiating with respect to ω_b yields

$$D_{m^*} \approx \frac{g^2}{\Delta_{bc}^2} - \frac{\Omega_0^2}{\Delta_{m^*}^2}. \quad (16)$$

Setting $D_{m^*} = 0$ gives the sweet-spot condition

$$\Delta_{m^*} = \pm \left| \frac{\Delta_{bc}}{g} \Omega_0 \right| \iff \Delta_{bd} = \pm \left| \frac{\Delta_{bc}}{g} \Omega_0 \right| - m^* \chi. \quad (17)$$

This prediction is confirmed by Fig. 2(a), where two sweet spots appear as symmetric dips on either side of the resonance. Their characteristic detuning is consistent with Eq. (17), which predicts $|\Delta_{bd}| \sim (|\Delta_{bc}|/g)\Omega_0 \approx 0.5$ MHz for the parameters used here ($g/|\Delta_{bc}| \simeq 0.1$). Overall, the agreement between the analytics and exact numerics shows that in the selective regime a low-power drive cancels the bare Lamb-shift susceptibility.

c. Sweet-Spot Map and Regime Crossover. Fig. 2(c) shows $D_{0\rightarrow 1}$ as a function of drive amplitude Ω_0 and detuning Δ_{bd} . The key point is that the selective and unselective sweet spots are not distinct phenomena, but rather different parts of a continuous contour defined by $D_{0\rightarrow 1} = 0$. At small Ω_0 , the selective-regime condition in Eq. (17) predicts the linear scaling $|\Delta_{bd}| \propto \Omega_0$, visible as two symmetric sweet spots on either side of the effective resonance. As Ω_0 increases, the inner branch, closer to resonance and at smaller $|\Delta_{bd}|$, enters a nonperturbative

regime in which the weak-drive approximation underlying Eq. (8) no longer applies. The remaining perturbative branch then crosses over toward the unselective regime and approaches the cubic scaling predicted by Eq. (14). At still larger Ω_0 , the Stark-shift contribution broadens the detuning range over which $D_{0 \rightarrow 1} \approx 0$, improving robustness against flux noise by maintaining susceptibility cancellation over a wider range of fluctuations, albeit closer to the limits of perturbation theory. Higher-order drive effects generate additional sweet spots; see App. E. The same mechanism is not restricted to the FTT and is illustrated for a SNAIL-based circuit in App. F. Finally, although the sweet-spot condition does not in general suppress sensitivity to drive-amplitude noise, for AWG noise levels typical of experiments the associated dephasing is negligible, see App. G.

III. OVERALL DEPHASING RATE

A periodic drive mitigates low-frequency flux-induced dephasing by coupling the FTT states and suppressing the first-order sensitivity of the dressed cavity frequency to fluctuations of ω_b . However, the drive also contributes new dephasing channels when FTT relaxation is present.

To quantify the total cavity dephasing rate, the system is modeled with a stochastic master equation. For a given flux-noise realization $\delta\Phi(t)$, the density matrix $\rho(t)$ obeys

$$\dot{\rho} = -i [H_{\text{sys}} + \delta\omega_b(t) b^\dagger b, \rho] + \gamma_{b\downarrow} \mathcal{D}[b]\rho, \\ H_{\text{sys}} = H_{\text{static}} + 2\Omega_0 \cos(\omega_d t) (b + b^\dagger). \quad (18)$$

Here $\mathcal{D}[b]\rho = b\rho b^\dagger - \frac{1}{2}\{b^\dagger b, \rho\}$ describes FTT relaxation at rate $\gamma_{b\downarrow}$ (the FTT energy-relaxation rate).

Flux noise in these devices typically exhibits a $1/f$ spectrum. The corresponding power spectral density is taken as

$$S_{1/f}(\omega) = \int_{-\infty}^{\infty} d\tau \langle \delta\Phi(\tau) \delta\Phi(0) \rangle e^{i\omega\tau} = \frac{S_0^2}{|\omega/2\pi|}, \quad (19)$$

where S_0 sets the flux-noise amplitude.

The following analysis is carried out in the effective-drive frame to separate the roles of relaxation and dephasing. The unselective and selective regimes are treated in turn.

A. Total dephasing in the unselective regime

In the unselective regime $|\Delta_{bd}| \gg |\chi|$, the master equation can be expanded to first order in $|\chi/\Delta_{bd}|$ about the sweet-spot condition (Eq. (14)). This yields the Lindblad master equation for the density matrix in the effective drive frame ρ_d ,

$$\dot{\rho}_d = -i[H_d + H_{\text{noise}}^d, \rho_d] + \tilde{\gamma}_{b\downarrow}^{\text{un}} \mathcal{D}[\sigma^-]\rho_d + \gamma_{b\uparrow}^{\text{un}} \mathcal{D}[\sigma^+]\rho_d \\ + 2\gamma_{c\phi}^{\text{un}} \mathcal{D}[c^\dagger c]\rho_d + \tilde{\gamma}_{c\downarrow}^{\text{un}} \mathcal{D}[c]\rho_d, \quad (20)$$

See App. H 4 for the derivation. The effective rates $\tilde{\gamma}_{b\downarrow}^{\text{un}}$, $\gamma_{b\uparrow}^{\text{un}}$, $\gamma_{c\phi}^{\text{un}}$, and $\tilde{\gamma}_{c\downarrow}^{\text{un}}$ are given below. Here, H_d is the effective driven Hamiltonian Eq. (11), and H_{noise}^d is the noise Hamiltonian in the effective-drive frame. In the unselective limit it takes the approximate form

$$H_{\text{noise}}^d \approx \delta\omega_b \left[\left(1 - \frac{\Delta_{bd}}{2K}\right) \sigma^+ \sigma^- + \frac{2g^2}{\Delta_{bc}^2} \left(1 - \frac{2K}{\Delta_{bc}}\right) \sigma^+ \sigma^- c^\dagger c \right. \\ \left. - \frac{1}{2} \sqrt{\frac{\Delta_{bd}}{K}} \sigma_x \right]. \quad (21)$$

Although the sweet-spot condition cancels the leading direct cavity-dephasing contribution ($D_{0 \rightarrow 1} = 0$), it does not eliminate all noise channels in the effective-drive frame, which are discussed below.

a. Interpretation of the dissipators. The dissipators in Eq. (20) represent (i) drive-renormalized FTT relaxation, $\tilde{\gamma}_{b\downarrow}^{\text{un}} \mathcal{D}[\sigma^-]$; (ii) drive-induced FTT excitation, $\gamma_{b\uparrow}^{\text{un}} \mathcal{D}[\sigma^+]$, which produces photon-shot-noise dephasing of the cavity via the dispersive coupling [37, 38]; (iii) cavity pure dephasing, $2\gamma_{c\phi}^{\text{un}} \mathcal{D}[c^\dagger c]$ (the prefactor 2 is conventional); and (iv) renormalized cavity decay, $\tilde{\gamma}_{c\downarrow}^{\text{un}} \mathcal{D}[c]$. At the sweet-spot condition of Eq. (14), the corresponding rates are

$$\tilde{\gamma}_{b\downarrow}^{\text{un}} = \gamma_{b\downarrow} \left(1 - \frac{\Delta_{bd}}{K}\right), \\ \gamma_{b\uparrow}^{\text{un}} = \frac{\gamma_{b\downarrow}}{16} \left(\frac{\Delta_{bd}}{K}\right)^2 \left(1 - \frac{2\Delta_{bd}}{K}\right), \\ \gamma_{c\phi}^{\text{un}} = \frac{1}{2} \gamma_{b\downarrow} \left(\frac{g}{\Delta_{bc}}\right)^4 \frac{K}{\Delta_{bd}}, \quad \tilde{\gamma}_{c\downarrow}^{\text{un}} = \gamma_{c\downarrow} \left(1 + \frac{g^2}{\Delta_{bc}^2}\right). \quad (22)$$

The cavity-decay renormalization scales as g^2/Δ_{bc}^2 and remains small (e.g., ~ 0.01) even in the relatively strong dispersive-coupling regime.

b. Interpretation of the noise Hamiltonian. At the sweet spot, cavity dephasing $c^\dagger c$ in Eq. (21) vanishes by construction, suppressing direct cavity-frequency noise. Also, AC-Stark-shift-induced cavity-frequency fluctuations are neglected because the FTT is barely excited if it is initialized in the ground state. Residual dephasing is dominated by a transverse component $\propto \sigma_x$, which induces FTT depolarization under $1/f$ flux noise. The transverse-noise-induced excitation rate is

$$\gamma_{b\uparrow}^{1/f, \text{un}} = \left(\frac{\Omega_0}{\Delta_{bd}} \frac{\partial \omega_b}{\partial \Phi}\right)^2 \frac{S_0^2}{|\Delta_{bd}/2\pi|} \stackrel{(14)}{=} \frac{1}{4} \left(\frac{\partial \omega_b}{\partial \Phi}\right)^2 \frac{S_0^2}{|K/2\pi|} \quad (23)$$

at the sweet-spot condition.

c. Total cavity dephasing rate. In the unselective regime, evaluated at the sweet spot, the total cavity de-

phasing rate is the sum of the three contributions in

$$\begin{aligned}\Gamma_{\phi,\text{total}} &= \gamma_{b\uparrow}^{1/f,\text{un}} + \gamma_{b\uparrow}^{\text{un}} + \gamma_{c\phi}^{\text{un}} \\ &= \left(\frac{\partial\omega_b}{\partial\Phi}\right)^2 \frac{S_0^2}{4K} + \frac{\gamma_{b\downarrow}}{16} \left(\frac{\Delta_{bd}}{K}\right)^2 \left(1 - \frac{2\Delta_{bd}}{K}\right) \\ &\quad + \frac{\gamma_{b\downarrow}}{2} \left(\frac{g}{\Delta_{bc}}\right)^4 \frac{K}{\Delta_{bd}}.\end{aligned}\quad (24)$$

We approximate the photon-shot-noise contribution to cavity dephasing by the FTT heating rate $\gamma_{b\uparrow}^{\text{un}}$ in the limit $\chi \gg \tilde{\gamma}_{b\downarrow}$ [39], because a single photon event is sufficient to entirely destroy phase coherence before the photon decays. Under this approximation, each contribution in Eq. (24) agrees well with numerical simulations, as verified in App. J. We also emphasize that the drive induces only a small correction to $\gamma_{c\downarrow}$, scaling as $g^2/\Delta_{bc}^2 \ll 1$; see App. J.

B. Total dephasing in the selective regime

Operating at the selective-regime sweet spot protects the cavity states $|g, m^*\rangle \rightarrow |g, m^* + 1\rangle$ from flux-noise dephasing. Under drive, however, the residual dephasing budget in the selective regime is set by two contributions: (i) cavity pure dephasing at a rate $\gamma_{c\phi}^{\text{sel}} = (g^2/\Delta_{bc}^2)\gamma_{b\downarrow}$, arising from the Purcell-like mixing of cavity and FTT decay channels, and (ii) FTT heating at a rate $\gamma_{b\uparrow}^{\text{sel}} \propto 1/|\Delta_{bd}|$ induced by photon shot noise (see App. H 4 for derivations). Because the shot-noise contribution scales unfavorably as $1/|\Delta_{bd}|$, operating at larger detuning—i.e., in the unselective regime—yields a lower total dephasing rate.

C. Numerical evaluation of the total dephasing rate

Full-system simulations are used to extract the cavity dephasing rate under drive; the numerical procedure and fitting protocol are summarized in App. I. The noise amplitude S_0 and the detuning Δ_{bd} are sampled over the parameter range of interest. For each Δ_{bd} , the drive amplitude is chosen to satisfy the sweet-spot condition, and the corresponding total dephasing rate $\Gamma_{\phi,\text{total}}$ is extracted.

Fig. 3(a) summarizes the results. For $S_0 = 10^{-5}/\Phi_0$, the drive-induced contribution $\gamma_{b\uparrow}^{1/f,\text{un}}$ dominates $\Gamma_{\phi,\text{total}}$ over the scanned detuning range. At large detuning, $|\Delta_{bd}| > |\chi| \simeq 2$ MHz, the total rate is well described by $\gamma_{b\uparrow}^{1/f,\text{un}}$, which is independent of the detuning. At smaller detuning, i.e. $-\Delta_{bd}/2\pi = 1$ MHz $< |\chi/2\pi|$, the rate increases as the dynamics enters the selective regime, consistent with the $1/|\Delta_{bd}|$ scaling.

For smaller noise amplitude, e.g. $S_0 = 10^{-6}/\Phi_0$, the S_0 -independent drive-induced contributions $\gamma_{b\uparrow}^{\text{un}}$ and $\gamma_{c\phi}^{\text{un}}$ dominate $\Gamma_{\phi,\text{total}}$. At large detuning ($-\Delta_{bd}/2\pi =$

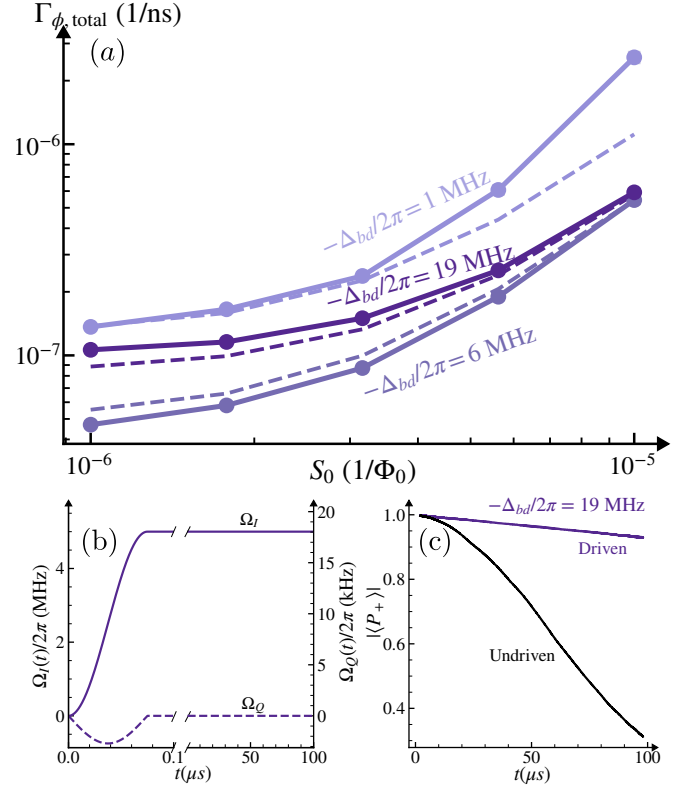


Figure 3. Numerical evaluation and time-domain validation of the total cavity dephasing rate under sweet-spot driving. (a) Total dephasing rate $\Gamma_{\phi,\text{total}}$ extracted from full-system simulations as a function of detuning Δ_{bd} , with the drive amplitude chosen at each detuning to satisfy the sweet-spot condition. S_0 denotes the amplitude of $1/f$ flux noise. (b) DRAG-style adiabatic ramp pulse. (c) Time-domain decay of $P_X(t) = \text{Tr}[\rho(t) X(t)]$, with $X(t) = |\phi_{g,1}(t)\rangle\langle\phi_{g,0}(t)| + \text{H.c.}$. The time-dependent Floquet states are labeled by their bare-state correspondence. The driven system exhibits a cavity dephasing time of ~ 2 ms, compared to ~ 100 μ s in the undriven case.

19 MHz), $\gamma_{b\uparrow}^{\text{un}} \propto (\Delta_{bd}/K)^2$ dominates while $\gamma_{c\phi}^{\text{un}} \sim K/\Delta_{bd}$ is suppressed; the opposite hierarchy holds at small detuning ($-\Delta_{bd}/2\pi = 1$ MHz). Consequently, $-\Delta_{bd}/2\pi = 6$ MHz yields the minimal $\Gamma_{\phi,\text{total}}$ within the scanned range. In the undriven case, the dephasing time is ~ 0.1 – 1 ms as S_0 varies from $10^{-5}/\Phi_0$ to $10^{-6}/\Phi_0$. With $\Delta_{bd}/2\pi \simeq -6$ MHz and sweet-spot driving, the dephasing time improves by up to $\sim 20\times$ across the considered S_0 range, confirming the overall dephasing enhancement.

To visualize the dephasing enhancement directly, we perform time-domain simulations that include relaxation and stochastic flux noise with $S_0 = 10^{-5}/\Phi_0$. The initial state is $|\psi_0\rangle = \frac{1}{\sqrt{2}}(|g, 0\rangle + |g, 1\rangle)$. With drive, the state is adiabatically mapped onto the target Floquet states using a shaped pulse $2\Omega_0(t) \cos(\omega t + \phi(t))$. In the rotating frame this yields the I and Q quadratures

$$\Omega_I(t) = \Omega(t) \cos \phi(t), \quad \Omega_Q(t) = \Omega(t) \sin \phi(t). \quad (25)$$

We implement a DRAG correction, $\Omega_Q(t) = -\dot{\Omega}_I(t)/\Delta_{bd}$, which accelerates the adiabatic transfer relative to $\Omega_I(t)$ -only ramps (see App. K). A pulse ramp with $\tau = 75$ ns is shown in Fig. 3(b).

The dephasing is quantified by the coherence operator

$$X(t) = |\phi_{g,1}(t)\rangle\langle\phi_{g,0}(t)| + \text{H.c.}, \quad (26)$$

$$P_X(t) = \text{Tr}[\rho(t) X(t)], \quad (27)$$

where $|\phi_{g,1}(t)\rangle$ and $|\phi_{g,0}(t)\rangle$ are adiabatically mapped from the bare states. Without drive, $1/f$ noise produces sub-Gaussian free-induction decay. At the driven sweet spot, the dominant residual channel is Markovian, arising from FTT depolarization, and yields the exponential decay observed in Fig. 3(c). The dephasing time T_2 is extracted from an exponential fit to $P_X(t)$ (see Appendix I). Under sweet-spot driving, the dephasing time reaches ~ 2 ms, compared with ~ 100 μ s in the undriven case.

IV. APPLICATION TO BOSONIC ENCODINGS

In the previous sections, we showed that operating at a driven sweet spot can extend the cavity dephasing time by more than an order of magnitude. Here we apply the same strategy to bosonic quantum error correction (QEC). The central observation is that, in the unselective regime, the drive does not resolve individual photon-number states, so a single choice of drive parameters simultaneously suppresses the dephasing rates of all relevant cavity transitions induced by FTT-frequency fluctuations. This property is precisely what is needed for bosonic encodings, which store quantum information in superpositions of multiple Fock states.

Throughout this section, we fix the drive amplitude at $\Omega_0/2\pi = 10$ MHz and sweep the detuning Δ_{bd} to identify the operating point that simultaneously minimizes the relevant dephasing rates. The metrics below—transition dephasing rates and logical-subspace coherence proxies—quantify the suppression of inherited dephasing rather than a full logical error rate, which would additionally require modeling the complete QEC cycle.

A. Binomial encoding

As a concrete example, consider the minimal binomial code [40]

$$|0_L\rangle = \frac{|0\rangle + |4\rangle}{\sqrt{2}}, \quad |1_L\rangle = |2\rangle, \quad (28)$$

the simplest binomial encoding that corrects a single photon-loss event. The drive is not intended to eliminate all dephasing; rather, it suppresses the specific transition-frequency fluctuations that dominate the logical-phase diffusion and recovery-sensitive coherences for this encoding. We now identify these transitions explicitly.

For the logical basis state $|0_L\rangle \propto |0\rangle + |4\rangle$, fluctuations of the $0 \rightarrow 4$ splitting, $\delta\tilde{\omega}_{c,0\rightarrow 4}(t)$, induce phase diffusion between $|0\rangle$ and $|4\rangle$ and therefore deform $|0_L\rangle$. Because this code does not protect against pure dephasing, the coherence of an arbitrary superposition $\alpha|0_L\rangle + \beta|1_L\rangle$ is carried by the bare-state coherences $|0\rangle \leftrightarrow |2\rangle$ and $|2\rangle \leftrightarrow |4\rangle$. Suppressing the corresponding dephasing rates $\gamma_{\phi,0\rightarrow 2}$ and $\gamma_{\phi,2\rightarrow 4}$ therefore directly improves logical coherence.

After a single photon loss, the code space maps as $|0_L\rangle \rightarrow |3\rangle$ and $|1_L\rangle \rightarrow |1\rangle$, so the logical information temporarily resides in the $|1\rangle \leftrightarrow |3\rangle$ coherence prior to correction. Suppressing $\gamma_{\phi,1\rightarrow 3}$ is thus essential for high-fidelity recovery. During the recovery itself, intermediate superpositions populate the $\{|0\rangle, |3\rangle, |4\rangle\}$ manifold and the $\{|1\rangle, |2\rangle\}$ manifold, making the corresponding transition dephasing rates—e.g., $\gamma_{\phi,0\rightarrow 3}$, $\gamma_{\phi,3\rightarrow 4}$, and $\gamma_{\phi,1\rightarrow 2}$ —additional targets for suppression.

To quantify the improvement, we sweep the detuning Δ_{bd} at fixed $\Omega_0/2\pi = 10$ MHz and evaluate the dephasing rate due to flux noise,

$$\gamma_{\phi,n\rightarrow m} = S_0 \left| \frac{\partial \tilde{\omega}_{c,n\rightarrow m}}{\partial \Phi} \right| \sqrt{2 \ln(|\omega_{\text{ir}} t|)}. \quad (29)$$

Here, ω_{ir} is the infrared cutoff of the $1/f$ noise spectrum (set by the inverse of the total measurement time) and t is the experimental timescale. An earlier experiment reported $\ln(|\omega_{\text{ir}} t|) \approx 4$ [41], and we use this value in the following numerics. The results are shown in Fig. 4(a). The minima of several dephasing rates overlap near $-\Delta_{bd}/2\pi \simeq 30$ MHz, consistent with the unselective regime $\Delta_{bd} \gg |\chi|$ (here $|\chi|/2\pi \sim 0.5$ MHz).

Choosing $\Delta_{bd}/2\pi = -30$ MHz yields a worst-case reduction of the dephasing rate by a factor of ~ 13 across the relevant transitions. This improvement is confirmed by Wigner tomography [Fig. 4(b)], which serves as a direct proxy for the preservation of coherence within the logical subspace. For the initial state $|+_L\rangle = (|0_L\rangle + |1_L\rangle)/\sqrt{2}$, the undriven Wigner function reduces to a mixture of Fock states after 300 μ s, indicating complete loss of logical coherence. In the driven case, the characteristic interference fringes are largely retained, confirming that the drive preserves the intra-code-word coherence required by the binomial encoding.

B. Dual-rail encoding

Dual-rail encoding is widely used because the dominant single-photon-loss error maps to a detectable erasure [20, 42]. A single dual-rail qubit is encoded in two cavities with logical states $|0_L\rangle = |01\rangle$ and $|1_L\rangle = |10\rangle$. The fault-tolerance advantage of this encoding relies on the error hierarchy in which single-photon loss dominates over dephasing; inherited cavity dephasing from a flux-tunable coupler can degrade this bias. To implement two-qubit gates, one approach is to couple two dual-rail

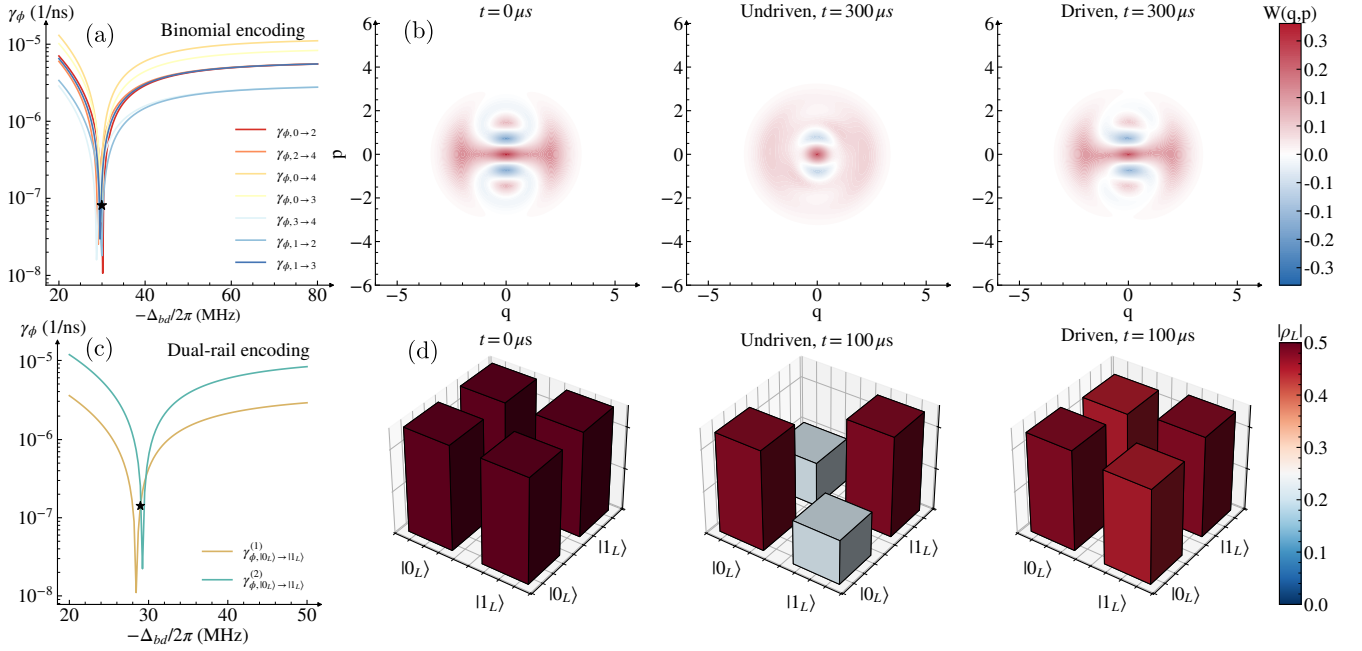


Figure 4. **Driven sweet-spot protection for bosonic encodings** ($\Omega_0/2\pi = 10$ MHz throughout). *Binomial code*—(a) Dephasing rate $\gamma_{\phi, n \rightarrow m}$ of cavity transitions due to FTT-frequency fluctuations as a function of Δ_{bd} ; the star marks $\Delta_{bd}/2\pi = -30$ MHz, where all relevant $\gamma_{\phi, n \rightarrow m}$ are suppressed by at least $\sim 13\times$. (b) Wigner function of $|+_L\rangle$ at $t = 300 \mu s$: the driven case retains the interference fringes that are lost without the drive. *Dual-rail encoding*—(c) Dephasing rate $\gamma_{\phi, 0 \rightarrow 1}^{(i)}$ for each cavity as a function of Δ_{bd} ; both cavities exhibit $\sim 10\times$ lower dephasing near $-\Delta_{bd}/2\pi \sim 29$ MHz. (d) Logical-subspace tomography for one dual-rail qubit: the drive suppresses dephasing, while leakage from single-photon loss remains unchanged.

qubits via an FTT and exploit its three-wave-mixing interaction.

In this architecture, the FTT (mode b) couples to two cavities labeled by i . Flux fluctuations $\delta\Phi(t)$ modulate the FTT frequency, and this modulation enters the cavity spectrum through the Lamb shift, producing cavity-frequency fluctuations

$$\delta\tilde{\omega}_c^{(i)}(t) = \left(\frac{g^{(i)}}{\Delta_{bc}^{(i)}} \right)^2 \delta\omega_b(t), \quad (30)$$

where $g^{(i)}$ and $\Delta_{bc}^{(i)}$ are the coupling strength and detuning between the FTT and cavity i , respectively. Because each cavity forms one rail of a dual-rail qubit, these fluctuations contribute directly to logical dephasing. By appropriately driving the coupler, the dephasing rates of both cavities due to inherited FTT-frequency noise can be reduced simultaneously.

Fig. 4(c) shows the numerically computed dephasing rate $\gamma_{\phi, 0 \rightarrow 1}^{(i)}$ for each cavity as Δ_{bd} is swept. Choosing $-\Delta_{bd}/2\pi \sim 29$ MHz yields $\sim 10\times$ lower dephasing rates for both cavities than in the undriven case. We then simulate the dynamics of a single dual-rail qubit and show the state tomography in the logical subspace in Fig. 4(d). The simulation confirms that the drive largely suppresses pure dephasing within the logical subspace, while single-photon-loss-induced leakage out of the logical subspace

remains nearly unchanged between the driven and undriven cases. This demonstrates that the driven sweet spot acts as a bias-preserving knob: it mitigates inherited dephasing while leaving the dominant single-photon-loss channel—and hence the favorable error hierarchy of the dual-rail encoding—intact.

V. CONCLUSION

In summary, we have introduced a hardware-efficient protocol to dynamically suppress cavity dephasing caused by flux noise inherited from a coupled nonlinear element. By applying a continuous, off-resonant microwave drive to the FTT, we generate a photon-number-dependent AC-Stark shift that cancels the cavity frequency's first-order susceptibility to coupler-frequency fluctuations. Our theoretical analysis and Monte Carlo simulations show that operating at this driven sweet spot can extend the cavity dephasing time by more than an order of magnitude, achieving a $\sim 20\times$ improvement from $\sim 100 \mu s$ to ~ 2 ms.

While the continuous drive inevitably introduces secondary decoherence channels, such as photon-shot-noise dephasing from drive-induced FTT excitation, we show that operating in the unselective regime at larger detunings parametrically suppresses these effects. In this

regime, the primary flux-noise cancellation dominates, resulting in a net enhancement of coherence. Crucially, this mitigation strategy is highly hardware-efficient: it circumvents the need for fast flux tuning during idling, thereby avoiding unwanted transitions that can occur when the coupler frequency is swept past cavity resonances.

Furthermore, we have shown that the mechanism is not limited to the FTT: because it is rooted in drive-induced energy shifts, it extends to other flux-dependent nonlinear elements such as SNAIL-based couplers. We also showed that the protocol is directly relevant for bosonic quantum error correction, where a single drive setting can simultaneously suppress the flux sensitivity of multiple critical transitions in binomial and dual-rail encodings. Overall, the approach provides a practical route toward preserving the noise bias needed for cavity-based QEC at non-sweet-spot flux operating points.

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Appendix A: Hamiltonian Derivation and Noise Analysis

This appendix provides an analytical approximation showing that, to leading order, the dominant effect of flux noise is a fluctuation of the FTT frequency.

The full Hamiltonian of the system shown in Fig. 1(a) is

$$H_{\text{full}}(t) = 4E_C(n - n_g)^2 - E_{J,\text{eff}}(\Phi + \delta\Phi(t)) \cos[\varphi - \varphi_0(\Phi + \delta\Phi(t))] + \omega_c c^\dagger c + g'n(c^\dagger + c) + 2\Omega'_0 \cos(\omega_d t) n. \quad (\text{A1})$$

Here E_C is the charging energy, n is the charge operator, and n_g is the offset charge. The operator c annihilates photons in the cavity mode of frequency ω_c . The charge-charge coupling strength is g' , and the final term describes a charge drive applied to the FTT with amplitude Ω'_0 and frequency ω_d .

Both the effective Josephson energy $E_{J,\text{eff}}$ and the potential phase φ_0 depend on flux:

$$E_{J,\text{eff}}(\Phi + \delta\Phi(t)) = (E_{J1} + E_{J2}) \cos\left[\frac{\pi(\Phi + \delta\Phi(t))}{\Phi_0}\right],$$

$$\varphi_0(\Phi + \delta\Phi(t)) = d \tan\left[\frac{\pi(\Phi + \delta\Phi(t))}{\Phi_0}\right], \quad (\text{A2})$$

where E_{J1} and E_{J2} are the Josephson energies of the two junctions, $\Phi_0 = h/2e$ is the superconducting flux quantum, and $d \equiv (E_{J2} - E_{J1})/(E_{J1} + E_{J2})$ is the SQUID-loop asymmetry.

Flux noise in these systems is typically observed to follow a $1/f$ characteristic. Accordingly, the power spectral density of these Gaussian fluctuations is

$$S_{1/f}(\omega) = \int_0^t \langle \delta\Phi(0) \delta\Phi(\tau) \rangle e^{i\omega\tau} d\tau = \frac{A_0^2}{|\omega/2\pi|}, \quad (\text{A3})$$

where A_0 is the flux-noise spectral-density amplitude.

For small flux fluctuations $\delta\Phi(t) \ll \Phi_0$, a Taylor ex-

pansion of H_{full} gives

$$H_{\text{full}}(\Phi + \delta\Phi(t)) \approx H_{\text{sys}}(\Phi) + \underbrace{\delta\Phi(t) \frac{\partial H_{\text{sys}}}{\partial \Phi}}_{H'_{\text{noise}}} + \mathcal{O}((\delta\Phi)^2), \quad (\text{A4})$$

where the system Hamiltonian is evaluated at the static bias point Φ ,

$$H_{\text{sys}}(\Phi) = 4E_C(n - n_g)^2 - E_{J,\text{eff}}(\Phi) \cos[\varphi - \varphi_0(\Phi)] + \omega_c c^\dagger c + g'n(c^\dagger + c) + 2\Omega'_0 \cos(\omega_d t) n. \quad (\text{A5})$$

The dependence on Φ is dropped below for compactness.

In the transmon regime ($E_C \ll E_{J,\text{eff}}$), the wavefunction is localized in the potential well, and the Hamiltonian can be expressed using ladder operators:

$$\varphi - \varphi_0 = \varphi_{\text{ZPF}} (b + b^\dagger),$$

$$n = \frac{-i(b - b^\dagger)}{2\varphi_{\text{ZPF}}}, \quad (\text{A6})$$

$$\varphi_{\text{ZPF}} = \left(\frac{2E_C}{E_{J,\text{eff}}(\Phi)} \right)^{\frac{1}{4}}.$$

Expanding the cosine potential to fourth order and applying the rotating wave approximation (RWA) yields

$$H_{\text{sys}}(t) \approx \omega_b b^\dagger b + \frac{K}{2} b^{\dagger 2} b^2 + \omega_c c^\dagger c + g(b + b^\dagger)(c + c^\dagger) + 2\Omega_0 \cos(\omega_d t)(b + b^\dagger), \quad (\text{A7})$$

where the FTT frequency, anharmonicity, renormalized drive amplitude and coupling strength are $\omega_b = \sqrt{8E_C E_{J,\text{eff}}} - E_C$, $K = -E_C$, $\Omega_0 = \Omega'_0/2\varphi_{\text{ZPF}}$ and $g = g'/2\varphi_{\text{ZPF}}$ respectively.

The noise Hamiltonian in the oscillator basis defined at the static flux bias Φ is

$$\begin{aligned} H'_{\text{noise}}(t) &= \delta\Phi(t) \frac{\partial H_{\text{sys}}}{\partial \Phi} \\ &= -\delta\Phi(t) \left[\frac{\partial E_{J,\text{eff}}}{\partial \Phi} \cos(\varphi - \varphi_0) \right. \\ &\quad \left. + E_{J,\text{eff}} \frac{\partial \varphi_0}{\partial \Phi} \sin(\varphi - \varphi_0) \right] \\ &\approx -\delta\Phi(t) \left[\frac{\partial E_{J,\text{eff}}}{\partial \Phi} \left(1 - \frac{(\varphi - \varphi_0)^2}{2} \right) \right. \\ &\quad \left. + E_{J,\text{eff}} \frac{\partial \varphi_0}{\partial \Phi} (\varphi - \varphi_0) + \mathcal{O}((\varphi - \varphi_0)^3) \right] \\ &\approx -\delta\Phi(t) \left[\frac{\partial E_{J,\text{eff}}}{\partial \Phi} \left(1 - \frac{\varphi_{\text{ZPF}}^2}{2} (b + b^\dagger)^2 \right) \right. \\ &\quad \left. + E_{J,\text{eff}} \frac{\partial \varphi_0}{\partial \Phi} \varphi_{\text{ZPF}} (b + b^\dagger) \right]. \end{aligned} \quad (\text{A8})$$

This expression contains a term linear in $(b + b^\dagger)$ and a quadratic contribution proportional to $(b + b^\dagger)^2$. The linear term drives transitions near the oscillator frequency ω_b , with rates governed by $S_{1/f}(\omega_b)$. Expanding $(b + b^\dagger)^2 = 2b^\dagger b + 1 + b^2 + b^{\dagger 2}$ shows that the squeezing terms (b^2 and $b^{\dagger 2}$) drive transitions near $2\omega_b$, with rates proportional to $S_{1/f}(2\omega_b)$. Because $S_{1/f}(\omega)$ is strongly suppressed at these high frequencies ($\omega_b/2\pi \sim \text{GHz}$) compared to the low-frequency noise ($\omega/2\pi \rightarrow 0$ Hz) responsible for pure dephasing, both the linear and squeezing transition channels are neglected. Retaining only the number-conserving contribution ($\propto b^\dagger b$) from the quadratic term gives

$$H'_{\text{noise}}(t) \approx \frac{\partial E_{J,\text{eff}}}{\partial \Phi} \varphi_{\text{ZPF}}^2 \delta\Phi(t) b^\dagger b, \quad (\text{A9})$$

up to an additive constant. Using

$$\frac{\partial \omega_b}{\partial \Phi} = \frac{\partial \omega_b}{\partial E_{J,\text{eff}}} \frac{\partial E_{J,\text{eff}}}{\partial \Phi} = \varphi_{\text{ZPF}}^2 \frac{\partial E_{J,\text{eff}}}{\partial \Phi}, \quad (\text{A10})$$

one obtains

$$H'_{\text{noise}}(t) \approx \frac{\partial \omega_b}{\partial \Phi} \delta\Phi(t) b^\dagger b, \quad (\text{A11})$$

which is the noise Hamiltonian in Eq. (1).

Appendix B: Obtain H_d by a Schrieffer–Wolff transformation

Starting from the rotating-frame Hamiltonian (three-level truncation of the FTT),

$$\begin{aligned} H_R &= \Delta_{bd} |e\rangle\langle e| + (2\bar{\omega}_b + K) |f\rangle\langle f| + \bar{\omega}_c c^\dagger c \\ &\quad + \chi(|e\rangle\langle e| + 2|f\rangle\langle f|) c^\dagger c \\ &\quad + \Omega_0 (|g\rangle\langle e| + \sqrt{2}|e\rangle\langle f| + \text{h.c.}), \end{aligned} \quad (\text{B1})$$

the weak-drive regime $\Omega_0 \ll K$ allows the decomposition

$$H_R = H_0 + V, \quad (\text{B2})$$

where H_0 is the Ω_0 -independent part of Eq. (B1) (diagonal in the bare basis) and V is the drive term.

A Schrieffer–Wolff transformation uses an anti-Hermitian generator $D_d = -D_d^\dagger$ to obtain

$$H'_d = e^{-D_d} H_R e^{D_d}, \quad (\text{B3})$$

which is diagonal in the bare basis. To first order,

$$\begin{aligned} D_d &= \sum_m \left[-\frac{\Omega_0}{\Delta_m} |e, m\rangle\langle g, m| + \frac{\sqrt{2}\Omega_0}{\Delta_m + K} |f, m\rangle\langle e, m| - \text{h.c.} \right] \\ &\approx \sum_m \left[-\frac{\Omega_0}{\Delta_m} |e, m\rangle\langle g, m| + \frac{\sqrt{2}\Omega_0}{K} |f, m\rangle\langle e, m| - \text{h.c.} \right], \end{aligned} \quad (\text{B4})$$

where the approximation keeps the leading order in $\mathcal{O}(\Delta_m/K)$.

Keeping terms up to second order in the SWT gives

$$\begin{aligned} H'_d &\approx H_R + \sum_m \Omega_0^2 \left[-\frac{1}{\Delta_m} |g, m\rangle\langle g, m| + \left(\frac{1}{\Delta_m} \right. \right. \\ &\quad \left. \left. - \frac{1}{K} \right) |e, m\rangle\langle e, m| + \frac{1}{K} |f, m\rangle\langle f, m| \right]. \end{aligned} \quad (\text{B5})$$

Because the FTT is initialized in $|g\rangle$ and $\Omega_0 \ll K$, population of $|f\rangle$ remains negligible; projecting onto the $\{|g\rangle, |e\rangle\}$ subspace yields

$$\begin{aligned} H_d &= \tilde{\Delta}_{bd} \sigma^+ \sigma^- + \bar{\omega}_c c^\dagger c - \sum_m \frac{\Omega_0^2}{\Delta_m} |m\rangle\langle m| \\ &\quad + \sigma^+ \sigma^- \left[\chi c^\dagger c + \sum_m \frac{2\Omega_0^2}{\Delta_m} |m\rangle\langle m| \right], \end{aligned} \quad (\text{B6})$$

where $\tilde{\Delta}_{bd} = \Delta_{bd} - \frac{\Omega_0^2}{K}$ is the shifted FTT frequency.

Appendix C: Floquet Formalism

In this Appendix, we justify the use of the static rotating-frame Hamiltonian H_R [Eq. (7)] to approximate the quasi-energies and Floquet states of the driven dispersive Hamiltonian.

1. Floquet Theorem

Consider a Hilbert space $\mathcal{H}$ of dimension N and a time-dependent Hamiltonian $H(t)$ with period τ , satisfying $H(t + \tau) = H(t)$. According to Floquet's theorem, there exists a τ -periodic orthonormal basis $\{|\psi_n(t)\rangle\}$ and a set of quasienergies $\{\varepsilon_n\}$ such that the solutions to the Schrödinger equation, $|\phi_n(t)\rangle$, can be written as:

$$|\phi_n(t)\rangle = e^{-i\varepsilon_n t} |\psi_n(t)\rangle. \quad (C1)$$

2. Construction of the Solution

To acquire a differential equation that relates the periodic modes $|\psi_n(t)\rangle$ and quasienergies ε_n to the Hamiltonian $H(t)$, we substitute the ansatz solution into the time-dependent Schrödinger equation:

$$i \frac{\partial}{\partial t} |\Psi(t)\rangle = H(t) |\Psi(t)\rangle. \quad (C2)$$

Substituting $|\Psi(t)\rangle = e^{-i\varepsilon_n t} |\psi_n(t)\rangle$ yields the Floquet eigenvalue equation:

$$(H(t) - i\partial_t) |\psi_n(t)\rangle = \varepsilon_n |\psi_n(t)\rangle. \quad (C3)$$

Since the quantities on both sides of Eq. (C3) are periodic, we can treat the time dependence using Fourier analysis. A concise way to formalize this is to introduce the *Extended Hilbert space* $\mathcal{F}$, also known as Sambe space:

$$\mathcal{F} = \text{span}\{|\mu, n\rangle \mid \mu \in \mathbb{Z}, n = 1, \dots, N\}. \quad (C4)$$

We use the double ket notation $|\cdot\rangle\rangle$ to denote a basis vector in this extended space. Each basis vector corresponds to a function mapping time $t \in \mathbb{R}$ to a state in the physical space $\mathcal{H}$:

$$|\mu, n\rangle\rangle : \mathbb{R} \rightarrow \mathcal{H}. \quad (C5)$$

Specifically, these basis states are defined as:

$$|\mu, n\rangle\rangle(t) = |n\rangle e^{i\mu\omega_d t}, \quad \omega_d \equiv \frac{2\pi}{\tau}, \quad (C6)$$

where $\{|n\rangle\}$ is an orthonormal basis in $\mathcal{H}$. The inner product on $\mathcal{F}$ is defined by averaging over one period:

$$\begin{aligned} \langle\langle \mu', n' | \mu, n \rangle\rangle &\equiv \frac{1}{\tau} \int_0^\tau [|\mu', n'\rangle\rangle(t)]^\dagger |\mu, n\rangle\rangle(t) dt \\ &= \frac{1}{\tau} \int_0^\tau e^{-i\mu'\omega_d t} \langle n' | n \rangle e^{i\mu\omega_d t} dt \\ &= \delta_{n, n'} \frac{1}{\tau} \int_0^\tau e^{i(\mu - \mu')\omega_d t} dt \\ &= \delta_{\mu, \mu'} \delta_{n, n'}. \end{aligned} \quad (C7)$$

To rewrite Eq. (C3) in the basis $|\mu, n\rangle\rangle$, we define the quasienergy operator $\bar{Q} : \mathcal{F} \rightarrow \mathcal{F}$ such that its action in the time domain is:

$$(\bar{Q} | f \rangle\rangle)(t) = (H(t) - i\partial_t) | f \rangle\rangle(t). \quad (C8)$$

We define the vector $|\psi_n\rangle\rangle \in \mathcal{F}$ corresponding to the periodic function $|\psi_n(t)\rangle$ as:

$$|\psi_n\rangle\rangle(t) = |\psi_n(t)\rangle. \quad (C9)$$

Consequently, Eq. (C3) is represented as an eigenvalue problem in the extended Hilbert space:

$$\bar{Q} |\psi_n\rangle\rangle = \varepsilon_n |\psi_n\rangle\rangle. \quad (C10)$$

To compute the eigenvalues and eigenvectors, we find the matrix representation of $\bar{Q}$ in the basis $|\mu, n\rangle\rangle$:

$$\bar{Q} = \sum_{\mu', n'} \sum_{\mu, n} \bar{q}_{\mu' n', \mu n} |\mu', n'\rangle\rangle \langle\langle \mu, n |, \quad (C11)$$

with the corresponding matrix elements:

$$\begin{aligned} \bar{q}_{\mu' n', \mu n} &= \langle\langle \mu', n' | \bar{Q} | \mu, n \rangle\rangle \\ &= \frac{1}{\tau} \int_0^\tau [|\mu', n'\rangle\rangle(t)]^\dagger (\bar{Q} |\mu, n\rangle\rangle(t)) dt. \end{aligned} \quad (C12)$$

Substituting the explicit form $|\mu, n\rangle\rangle(t) = |n\rangle e^{i\mu\omega_d t}$:

$$\begin{aligned} \bar{q}_{\mu' n', \mu n} &= \frac{1}{\tau} \int_0^\tau e^{-i\mu'\omega_d t} \langle n' | (H(t) - i\partial_t) (|n\rangle e^{i\mu\omega_d t}) dt \\ &= \frac{1}{\tau} \int_0^\tau \langle n' | H(t) | n \rangle e^{i(\mu - \mu')\omega_d t} dt + \mu\omega_d \delta_{\mu', \mu} \delta_{n', n} \\ &= h_{n' n}^{(\mu - \mu')} + \mu\omega_d \delta_{\mu', \mu} \delta_{n', n}, \end{aligned} \quad (C13)$$

where $h_{n' n}^{(k)}$ are the Fourier components of the Hamiltonian:

$$H^{(k)} = \frac{1}{\tau} \int_0^\tau H(t) e^{ik\omega_d t} dt, \quad h_{n' n}^{(k)} = \langle n' | H^{(k)} | n \rangle. \quad (C14)$$

Thus, the operator $\bar{Q}$ can be written as a block matrix with indices μ running (for example) from -2 to 2 :

$$\bar{q} = \begin{pmatrix} \ddots & & & & & \\ & h^{(0)} - 2\omega_d I & h^{(1)} & h^{(2)} & h^{(3)} & h^{(4)} \\ & h^{(-1)} & h^{(0)} - \omega_d I & h^{(1)} & h^{(2)} & h^{(3)} \\ & h^{(-2)} & h^{(-1)} & h^{(0)} & h^{(1)} & h^{(2)} \\ & h^{(-3)} & h^{(-2)} & h^{(-1)} & h^{(0)} + \omega_d I & h^{(1)} \\ & h^{(-4)} & h^{(-3)} & h^{(-2)} & h^{(-1)} & h^{(0)} + 2\omega_d I \\ & & & & & \ddots \end{pmatrix}, \quad (\text{C15})$$

where I is the identity matrix. Note that the off-diagonal blocks depend on the difference in indices, $H^{(\mu-\mu')}$. Finally, the quasienergies are found by solving the characteristic equation:

$$\det(\bar{q} - \varepsilon \bar{I}) = 0, \quad (\text{C16})$$

where $\bar{I}$ is the identity operator in the extended Hilbert space.

3. Quasienergy and Floquet States under H_{disp}

In this subsection we relate an approximation of quasienergy and Floquet states to a rotating-wave approximation.

a. Quasienergy Operator for the Driven Dispersive Hamiltonian

The driven dispersive Hamiltonian is:

$$H(t) = \bar{\omega}_b b^\dagger b + \frac{K}{2} b^{\dagger 2} b^2 + \bar{\omega}_c c^\dagger c + \chi b^\dagger b c^\dagger c + 2\Omega_0 \cos(\omega_d t)(b + b^\dagger). \quad (\text{C17})$$

Following the recipe in the previous section, the quasienergy operator in the extended Hilbert space $\mathcal{F}$ is written as

$$\bar{Q} = \bar{Q}_0 + \bar{V}. \quad (\text{C18})$$

A convenient basis is $|\mu, (n, m)\rangle\rangle$, where $|\mu, (n, m)\rangle\rangle(t) = e^{i\mu\omega_d t}|n, m\rangle$ maps time to basis of the static dispersive Hamiltonian. The unperturbed component $\bar{Q}_0$ is diagonal in the basis $|\mu, (n, m)\rangle\rangle$:

$$\bar{Q}_0 = \sum_{\mu, n, m} \left(\bar{\omega}_b n + \frac{K}{2} n(n-1) + \bar{\omega}_c m + \chi n m + \mu \omega_d \right) |\mu, (n, m)\rangle\rangle \langle\langle \mu, (n, m) |, \quad (\text{C19})$$

$$\bar{V} = \Omega_0 \sum_{\mu, n, m} \sqrt{n+1} (|\mu+1, (n+1, m)\rangle\rangle \langle\langle \mu, (n, m) | + |\mu-1, (n+1, m)\rangle\rangle \langle\langle \mu, (n, m) |) + \text{h.c.} \quad (\text{C20})$$

The perturbation $\bar{V}$ arises from the periodic drive,

$$2\Omega_0 \cos(\omega_d t) = \Omega_0 (e^{i\omega_d t} + e^{-i\omega_d t}), \quad (\text{C21})$$

which mixes adjacent photon sectors. In the extended (Floquet) Hilbert space, the Fourier components $e^{\pm i\omega_d t}$ connect adjacent sectors $\mu \rightarrow \mu \mp 1$, while the drive excites FTT from $n \rightarrow n+1$. These processes have detunings

$$\Delta E_1 = (\bar{\omega}_b - \omega_d)n + (n-1)K + \chi m, \quad (\text{C22})$$

$$\Delta E_2 = (\bar{\omega}_b + \omega_d)n + (n-1)K + \chi m, \quad (\text{C23})$$

respectively. For our parameter regime of interest $|\bar{\omega}_b - \omega_d| \ll \omega_d$, we have parameters hierarchy

$$\Omega_0 < \Delta E_1 \ll \Delta E_2. \quad (\text{C24})$$

so that $\bar{V}$ can be treated perturbatively.

b. Approximation of Quasienergy and Floquet States

To show eigenvalue of H_R can approximate quasienergies, we block-diagonalize the operator $\bar{Q}$ via a Schrieffer-Wolff transformation within the near-resonant subspace selected by the hierarchy in Eq. (C24). Specifically, we define the projectors

$$\bar{P}_\mu = \sum_{n, m} |\mu - n, n, m\rangle \langle \mu - n, n, m|, \quad (\text{C25})$$

which collect the states coupled most strongly by the drive. Since $\bar{P}_\mu$ mixes different μ indices, it is convenient to relabel the basis with the unitary

$$\bar{R} = \sum_{\mu, n, m} |\mu, n, m\rangle \langle \mu - n, n, m|. \quad (\text{C26})$$

In this relabeled frame,

$$\bar{P}_\mu^R = \bar{R}^\dagger \bar{P}_\mu \bar{R} = \sum_{n,m} |\mu, n, m\rangle \langle \mu, n, m|, \quad (\text{C27})$$

$$\bar{Q}_R = \bar{R}^\dagger \bar{Q} \bar{R}, \quad (\text{C28})$$

so that $\bar{Q}_R$ acquires a block structure in the fixed- μ sectors:

$$\bar{q}_R = \begin{pmatrix} \ddots & \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \\ \cdots & [H_R - 2\omega_d I] & 0 & [b] & 0 & 0 & \cdots \\ \cdots & 0 & [H_R - \omega_d I] & 0 & [b] & 0 & \cdots \\ \cdots & [b^\dagger] & 0 & [H_R] & 0 & [b] & \cdots \\ \cdots & 0 & [b^\dagger] & 0 & [H_R + \omega_d I] & 0 & \cdots \\ \cdots & 0 & 0 & [b^\dagger] & 0 & [H_R + 2\omega_d I] & \cdots \\ \ddots & \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix}, \quad (\text{C29})$$

where $[O]$ denotes the matrix of O in the $\{|n, m\rangle\}$ basis. To first order, the effective quasienergy operator is given by the block-diagonal part,

$$\sum_\mu \bar{P}_\mu^R \bar{Q}_R \bar{P}_\mu^R. \quad (\text{C30})$$

It therefore suffices to diagonalize a single sector (e.g., $\mu = 0$), denoted by H_R :

$$H_R |\psi_j\rangle = E_j |\psi_j\rangle, \quad |\psi_j\rangle = \sum_{n,m} c_{nm}^j |n, m\rangle. \quad (\text{C31})$$

The quasienergy is then

$$\varepsilon_{j,\mu} = E_j + \mu\omega_d, \quad (\text{C32})$$

with corresponding Floquet states (in the relabeled frame)

$$|\psi_{j,\mu}\rangle_R = \sum_{n,m} c_{nm}^j |\mu, n, m\rangle. \quad (\text{C33})$$

c. Connection with the RWA

The relabeling $\bar{R}$ in extended Hilbert space implements the same rotation as a standard rotating-frame transformation in the physical Hilbert space. Acting on a Floquet basis state and evaluating its time dependence gives

$$\begin{aligned} (\bar{R}|\mu, n, m\rangle)(t) &= |\mu + n, n, m\rangle(t) = |n, m\rangle e^{i(\mu+n)\omega_d t} \\ &= (e^{in\omega_d t} |n, m\rangle) e^{i\mu\omega_d t}, \end{aligned} \quad (\text{C34})$$

so $\bar{R}$ amounts to multiplying $|n, m\rangle$ by the phase $e^{in\omega_d t}$. This is precisely the action of the rotating-frame unitary

$$U_R(t) = e^{i\omega_d t b^\dagger b}, \quad (\text{C35})$$

which yields the effective Hamiltonian

$$H_{\text{eff}}(t) = U_R^\dagger(t) H(t) U_R(t) - i U_R^\dagger(t) \dot{U}_R(t). \quad (\text{C36})$$

In this frame, terms oscillating at $\pm 2\omega_d$ are rapidly varying; neglecting them under the rotating-wave approximation produces a time-independent Hamiltonian H_R .

Appendix D: Exact calculation of $D_{0 \rightarrow 1}$

The driven circuit is simulated using the full system Hamiltonian H_{sys} in Eq. (A5), without further approximations. The parameters are $E_{J1} = E_{J2} = 30.19$ GHz, $E_C = 0.10$ GHz, $\omega_c/2\pi = 5.226$ GHz, $g'/2\pi = 0.050$ GHz, and $\Phi = 0.20 \Phi_0$. For these values, the effective parameters appearing in Eq. (1) are $\omega_b/2\pi = 6.16$ GHz, $K/2\pi = -0.10$ GHz, and $g/2\pi = 0.10$ GHz.

The basis is obtained by diagonalizing the static part of H_{sys} , i.e., $H_{\text{sys}}(t)$ with $\Omega_0 = \delta\Phi(t) = 0$. In the dispersive regime, these eigenstates are labeled by the corresponding bare basis states. The FTT (cavity) Hilbert space is truncated to 3 (4) levels. Increasing the truncation changes the results only marginally and does not affect the conclusions.

Floquet simulations are performed directly with $H_{\text{sys}}(t)$ to obtain time-periodic Floquet modes $|\phi_\alpha(t)\rangle$ and quasienergies ϵ_α , with $|\phi_\alpha(t+T)\rangle = |\phi_\alpha(t)\rangle$ and $T = 2\pi/\omega_d$. To assign each Floquet mode a label in a chosen bare basis $\{|j\rangle\}$, the time-averaged overlap weight

$$W_{\alpha j} = \frac{1}{T} \int_0^T dt |\langle j | \phi_\alpha(t) \rangle|^2 \quad (\text{D1})$$

is evaluated. Each Floquet mode α is labeled by the basis state $|j\rangle$ with the largest weight,

$$\text{label}(\alpha) = \arg \max_j W_{\alpha j}. \quad (\text{D2})$$

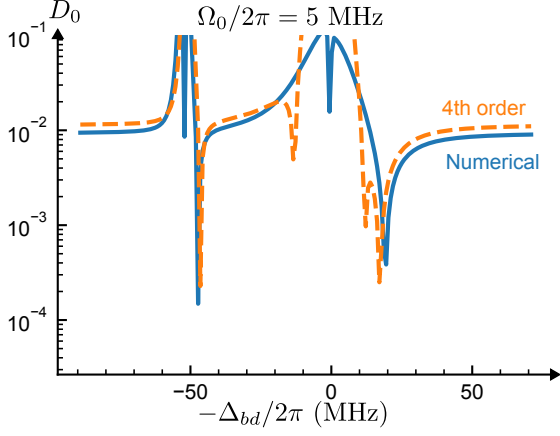


Figure 5. Susceptibility D_0 including higher-order corrections in the Schrieffer–Wolff expansion. An additional sweet spot appears near $-\Delta_{bd} \simeq K/2$, arising from hybridization between $|g\rangle$ and $|f\rangle$ states

This procedure yields a relabeling $\alpha \mapsto (n, m)$, where n and m denote the FTT and cavity excitation numbers, respectively, along with the corresponding quasienergies ϵ_{nm} .

Using the labeled quasienergies, the flux sensitivity of the $(g, 1) - (g, 0)$ quasienergy splitting is evaluated and normalized by the flux dispersion of ω_b . Specifically,

$$D_{0 \rightarrow 1} = \frac{\partial}{\partial \Phi} (\epsilon_{g1} - \epsilon_{g0}) \bigg/ \frac{\partial \omega_b}{\partial \Phi}, \quad (\text{D3})$$

which yields the exact $D_{0 \rightarrow 1}$ reported in Fig. 2.

Appendix E: Sweet spots from higher-order corrections

Higher-order, drive-induced terms in the Schrieffer–Wolff expansion modify the cavity susceptibility and can generate additional flux sweet spots. To illustrate this effect, we consider the vacuum transition frequency $\tilde{\omega}_{c,0 \rightarrow 1}$. Retaining terms up to fourth order yields the drive-induced correction

$$\begin{aligned} \tilde{\omega}_{c,0 \rightarrow 1} = & \bar{\omega}_c + \frac{\Omega_0^2}{-\Delta_{bd} - \chi} + \frac{\Omega_0^2}{\Delta_{bd}} - \frac{\Omega_0^4}{(-\Delta_{bd} - \chi)^3} \\ & + \frac{2\Omega_0^4}{(-\Delta_{bd} - \chi)^2(-2\Delta_{bd} - K - 2\chi)} - \frac{\Omega_0^4}{\Delta_{bd}^3} \\ & - \frac{2\Omega_0^4}{\Delta_{bd}^2(-2\Delta_{bd} - K)}. \end{aligned} \quad (\text{E1})$$

The resulting susceptibility D_0 is shown in Fig. 5. In addition to the sweet spots predicted at leading order, an extra sweet spot appears near $-\Delta_{bd} \simeq K/2$.

This feature originates from near-resonant hybridization between $|g, m\rangle$ and the higher FTT level $|f, m\rangle$. At

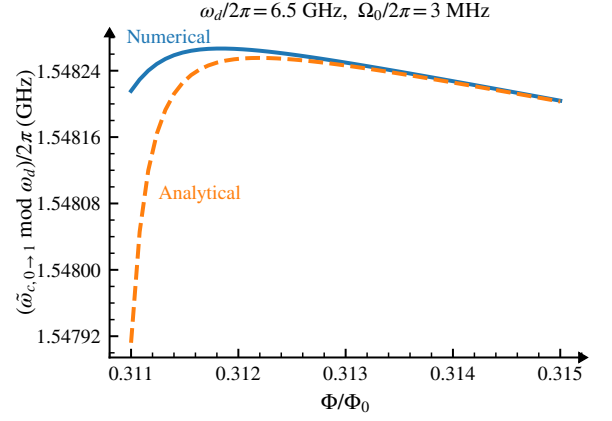


Figure 6. Drive-induced sweet spot in a SNAIL–cavity system. Under continuous drive, the cavity transition energy exhibits a maximum near $\Phi \simeq 0.311 \Phi_0$, indicating reduced first-order flux susceptibility. Solid lines show numerical results and dashed lines show the perturbative prediction from Eq. (9).

this detuning, the drive couples states separated by approximately the anharmonicity K , producing an additional Stark shift that alters the cancellation condition for D_0 . Operation at this sweet spot may therefore be more sensitive to drive-induced decoherence channels and should be assessed accordingly.

Appendix F: Example of a SNAIL coupler

The superconducting nonlinear asymmetric inductive element (SNAIL) is a three-wave-mixing circuit element widely used for parametric amplification and gate operations. When coupled capacitively to a cavity, the system Hamiltonian can be written as

$$\begin{aligned} H = & \omega_s s^\dagger s + \sum_j g_j (s + s^\dagger)^j + \omega_c c^\dagger c \\ & + g(s + s^\dagger)(c + c^\dagger) + 2\Omega_0 \cos(\omega_d t)(s + s^\dagger). \end{aligned} \quad (\text{F1})$$

Here ω_s is the SNAIL mode frequency, g_j are the nonlinear coefficients, g is the SNAIL–cavity coupling strength, s (c) is the SNAIL (cavity) mode operator, and ω_d is the drive frequency. The parameters ω_s , g_j , g , and Ω_0 are flux dependent through Φ .

To illustrate the generality of the drive-induced cancellation mechanism, the SNAIL is biased at $\Phi = 0.3 \Phi_0$, where the cubic nonlinearity g_3 enables three-wave mixing. At this operating point, $g_3/2\pi \approx 67$ MHz, $\omega_s/2\pi = 6.59$ GHz, and $\omega_c/2\pi = 5$ GHz.

A coherent drive with amplitude $\Omega_0 = 3$ MHz at $\omega_d = 6.5$ GHz produces a flux sweet spot in the cavity spectrum. As shown in Fig. 6, the cavity transition energy, extracted using the procedure in App. D, develops a local maximum near $\Phi \simeq 0.311 \Phi_0$, consistent with vanishing first-order flux susceptibility. The perturbative

expression in Eq. (9) reproduces this sweet spot quantitatively, demonstrating that the drive-induced cancellation mechanism extends beyond the FTT to more general nonlinear, flux-dependent modes.

Appendix G: Amplitude-noise-induced dephasing

With sweet spot drive, the net cavity frequency

$$\tilde{\omega}_{c,0 \rightarrow 1} = \bar{\omega}_c + \frac{\Omega_0^2 \chi}{\Delta_{bd}^2}. \quad (\text{G1})$$

is insensitive to flux noise, but it is sensitive to drive-amplitude noise.

We consider an amplitude-noise spectrum containing both white-noise and $1/f$ components,

$$S_\Omega(\omega) = (S_0^w)^2 + \frac{(S_0^{1/f})^2}{|\omega|/2\pi}, \quad (\text{G2})$$

where S_0^w and $S_0^{1/f}$ denote the amplitudes of the white-noise and $1/f$ noise contributions, respectively. Within leading-order perturbation theory, the corresponding cavity dephasing rate is

$$\frac{1}{T_\phi} = (S_0^w)^2 \left| \frac{\partial \tilde{\omega}_{c,0 \rightarrow 1}}{\partial \Omega} \right|^2 + S_0^{1/f} \left| \frac{\partial \tilde{\omega}_{c,0 \rightarrow 1}}{\partial \Omega} \right| \sqrt{2 \ln(|\omega_{\text{ir}} t|)}. \quad (\text{G3})$$

We evaluate this expression for a specific hardware platform to estimate the corresponding dephasing time. For the Zurich Instruments HDAWG, the white-noise floor is $25 \text{ nV}/\sqrt{\text{Hz}}$, and the $1/f$ corner frequency is approximately 100 kHz . These specifications imply a $1/f$ voltage-noise spectral density of approximately $7.9 \mu\text{V}/\sqrt{\text{Hz}}$ at 1 Hz .

To convert these voltage-noise specifications into Rabi-frequency noise amplitudes, we use the linear relation between the drive voltage V and the induced Rabi frequency Ω . Because cryogenic attenuation suppresses the signal and the amplitude noise by the same factor, it does not by itself change the fractional noise, so that $\delta\Omega/\Omega = \delta V/V$. Consequently, for a fixed target drive strength Ω , the effective drive-noise amplitudes depend on the AWG output voltage used to generate that drive. If the dominant instrument noise is set by an approximately additive voltage floor, then using a larger V_{AWG} and compensating with additional attenuation reduces the fractional noise $\delta V/V_{\text{AWG}}$. The white-noise and $1/f$ amplitudes are therefore

$$S_0^w = \Omega \times \frac{25 \text{ nV}/\sqrt{\text{Hz}}}{V_{\text{AWG}}}, \quad S_0^{1/f} = \Omega \times \frac{7.9 \mu\text{V}/\sqrt{\text{Hz}}}{V_{\text{AWG}}}, \quad (\text{G4})$$

where V_{AWG} is the peak output voltage of the instrument. As a representative operating point, for a target drive strength of $\Omega/2\pi = 10 \text{ MHz}$ and $V_{\text{AWG}} = 0.5 \text{ V}$, we obtain $S_0^w \approx 0.5\sqrt{\text{Hz}}$ and $S_0^{1/f} \approx 158\sqrt{\text{Hz}}$.

Using the parameters at the sweet spot in Fig. 2(a), we numerically evaluate

$$\left| \frac{\partial \tilde{\omega}_{c,0 \rightarrow 1}}{\partial \Omega} \right| = 0.044. \quad (\text{G5})$$

Assuming a typical measurement timescale such that $\sqrt{2 \ln(|\omega_{\text{ir}} t|)} \approx 4$ [41], we obtain a pure dephasing rate of

$$\frac{1}{T_\phi} \approx 35 \text{ s}^{-1}, \quad (\text{G6})$$

which corresponds to an amplitude-noise-limited dephasing time of

$$T_\phi \approx 29 \text{ ms}. \quad (\text{G7})$$

Appendix H: Transformation

This section derives H_d by transforming $H(t)$ and then expresses the master equation in Eq. (18) in the corresponding effective drive frame.

1. Transformation of Hamiltonian

In the regime $|g| \ll |\omega_c - \omega_b| \ll |\omega_c + \omega_b|$, counter-rotating terms can be neglected in the Hamiltonian H_{sys} (Eq. (18)), giving

$$H(t) \approx \omega_b b^\dagger b + \omega_c c^\dagger c + \frac{K}{2} b^{\dagger 2} b^2 + g(bc^\dagger + b^\dagger c) + 2\Omega_0 \cos(\omega_d t) (b + b^\dagger). \quad (\text{H1})$$

The static Hamiltonian H_{static} consists of the first four terms. To remove the interaction term $g(bc^\dagger + b^\dagger c)$, a second-order Schrieffer–Wolff transformation is applied with $U_{\text{disp}} = e^{D_{\text{disp}}}$,

$$H_{\text{static}}^{\text{disp}} = e^{-D_{\text{disp}}} H_{\text{static}} e^{D_{\text{disp}}}. \quad (\text{H2})$$

The generator D_{disp} is chosen to eliminate off-diagonal terms at first order,

$$D_{\text{disp}} = \sum_n \sqrt{n+1} \frac{g}{\omega_b + nK - \omega_c} |n+1\rangle \langle n| c - \text{h.c.} \quad (\text{H3})$$

Expanding in the detuning Δ_{bc} yields

$$D_{\text{disp}} \approx \frac{g}{\Delta_{bc}} b^\dagger c - \frac{gK}{\Delta_{bc}^2} (b^\dagger b - 1) b^\dagger c + \mathcal{O}((nK/\Delta_{bc})^2). \quad (\text{H4})$$

Applying the same transformation to the full (driven) Hamiltonian gives

$$\begin{aligned} H'_{\text{disp}} &= e^{-D_{\text{disp}}} H(t) e^{D_{\text{disp}}} \\ &\approx H_{\text{disp}} + \Omega_0 \cos(\omega_d t) \left[e^{-D_{\text{disp}}} (b + b^\dagger) e^{D_{\text{disp}}} \right] \\ &\approx H_{\text{disp}} + \Omega_0 \cos(\omega_d t) \left[b^\dagger + [D_{\text{disp}}, b] + \text{h.c.} \right], \end{aligned} \quad (\text{H5})$$

where $H_{\text{disp}} \equiv e^{-D_{\text{disp}}} H_{\text{static}} e^{D_{\text{disp}}}$. Using the expansion above,

$$[D_{\text{disp}}, b] \approx -\frac{g}{\Delta_{bc}} c - \frac{Kg}{\Delta_{bc}^2} b c^\dagger + \frac{2Kg}{\Delta_{bc}^2} b^\dagger b c. \quad (\text{H6})$$

These terms rotate at frequencies set by detunings involving the b - c manifold and the bare c mode. Because ω_d is far detuned from these transitions, the rotating-wave approximation justifies neglecting them, leaving

$$H'_{\text{disp}} \approx H_{\text{disp}} + \Omega_0 \cos(\omega_d t) (b + b^\dagger). \quad (\text{H7})$$

The explicit time dependence of the drive is removed by moving to the rotating frame defined by $U_R = e^{i\omega_d t b^\dagger b}$. The transformed Hamiltonian is

$$\begin{aligned} H'_R &= U_R^\dagger \left(H'_{\text{disp}} - i \frac{\partial}{\partial t} \right) U_R \\ &\approx H_R + (\Omega_0 e^{-i\omega_d t} b + \text{h.c.}), \end{aligned} \quad (\text{H8})$$

and a final RWA removes counter-rotating contributions at frequency $\pm 2\omega_d$, yielding the effective Hamiltonian H_R .

The subsequent transformation with U_d (Appendix B) maps H_R to H_d .

2. Transformation of H_{noise}

We consider the flux-noise Hamiltonian $H_{\text{noise}} = \frac{\partial \omega_b}{\partial \Phi} \delta \Phi(t) b^\dagger b$. After applying the dispersive transformation $U_{\text{disp}} = e^{D_{\text{disp}}}$, the number operator becomes

$$\begin{aligned} e^{-D_{\text{disp}}} b^\dagger b e^{D_{\text{disp}}} &= e^{-D_{\text{disp}}} b^\dagger e^{D_{\text{disp}}} e^{-D_{\text{disp}}} b e^{D_{\text{disp}}} \\ &\approx (b^\dagger + [D_{\text{disp}}, b^\dagger]) (b + [D_{\text{disp}}, b]) \\ &\approx b^\dagger b + [D_{\text{disp}}, b^\dagger b] + [D_{\text{disp}}, b^\dagger] [D_{\text{disp}}, b] \\ &\stackrel{(\text{H3})}{\approx} b^\dagger b + \frac{g^2}{\Delta_{bc}^2} c^\dagger c - \frac{4Kg^2}{\Delta_{bc}^3} b^\dagger b c^\dagger c \\ &\quad + \left(-\frac{g}{\Delta_{bc}} b c^\dagger + \frac{Kg}{\Delta_{bc}^2} b^\dagger b c^\dagger + \text{h.c.} \right). \end{aligned} \quad (\text{H9})$$

The non-number-conserving contributions (e.g., terms proportional to $b c^\dagger$ and $b^\dagger b c^\dagger$, together with their Hermitian conjugates) drive transitions at frequencies near $\Delta_{bc}/2\pi \sim \text{GHz}$. For $1/f$ flux noise, $S_{1/f}(\omega)$ is negligible at such frequencies, so these rapidly oscillating terms are discarded. The resulting dispersively transformed noise Hamiltonian is

$$H_{\text{noise}}^{\text{disp}} \approx \frac{\partial \omega_b}{\partial \Phi} \delta \Phi \left(b^\dagger b + \frac{g^2}{\Delta_{bc}^2} c^\dagger c - \frac{4Kg^2}{\Delta_{bc}^3} b^\dagger b c^\dagger c \right). \quad (\text{H10})$$

In the rotating frame defined by U_R , the remaining terms are diagonal in the relevant number operators, so the form is unchanged: $H_{\text{noise}}^R = H_{\text{noise}}^{\text{disp}}$. Under the drive

transformation $U_d = e^{D_d}$, the noise Hamiltonian is expanded to second order,

$$\begin{aligned} H_{\text{noise}}^{d'} &= U_d^\dagger H_{\text{noise}}^R U_d \\ &\approx H_{\text{noise}}^R + [D_d, H_{\text{noise}}^R] + \frac{1}{2} [D_d, [D_d, H_{\text{noise}}^R]], \end{aligned}$$

where we truncate the FTT Hilbert space to three levels in the regime of interest $|\Omega_0/K| \ll 1$,

$$\begin{aligned} H_{\text{noise}}^R &\approx \frac{\partial \omega_b}{\partial \Phi} \delta \Phi \left(|e\rangle \langle e| + 2|f\rangle \langle f| + \frac{g^2}{\Delta_{bc}^2} c^\dagger c \right. \\ &\quad \left. - \frac{4Kg^2}{\Delta_{bc}^3} (|e\rangle \langle e| + 2|f\rangle \langle f|) c^\dagger c \right). \end{aligned} \quad (\text{H11})$$

Evaluating the required commutators:

$$\begin{aligned} [D_d, b^\dagger b] &= -\sum_m \left[\frac{\Omega_0}{\Delta_m} |g, m\rangle \langle e, m| \right. \\ &\quad \left. + \frac{\sqrt{2}\Omega_0}{K} |e, m\rangle \langle f, m| + \text{h.c.} \right] \\ \frac{1}{2} [D_d, [D_d, b^\dagger b]] &= \sum_m \left[2 \left(\frac{\Omega_0}{\Delta_m} \right)^2 |g, m\rangle \langle g, m| \right. \\ &\quad \left. - 2 \left(\frac{\Omega_0}{\Delta_m} \right)^2 |e, m\rangle \langle e, m| - \frac{4\Omega_0^2}{K^2} |f, m\rangle \langle f, m| \right] \\ [D_d, c^\dagger c] &= 0 \end{aligned} \quad (\text{H12})$$

gives

$$\begin{aligned} H_{\text{noise}}^{d'} &\approx \frac{\partial \omega_b}{\partial \Phi} \delta \Phi \left(\frac{g^2}{\Delta_{bc}^2} c^\dagger c + \left(\frac{\Omega_0}{\Delta_m} \right)^2 |m\rangle \langle m| \right. \\ &\quad \left. + \sigma^+ \sigma^- - \sigma^- \sigma^+ - \left(\frac{4Kg^2}{\Delta_{bc}^3} c^\dagger c + 2 \left(\frac{\Omega_0}{\Delta_m} \right)^2 |m\rangle \langle m| \right) \right. \\ &\quad \left. - \sum_m \left[\frac{\Omega_0}{\Delta_m} \sigma^- |m\rangle \langle m| + \text{h.c.} \right] \right), \end{aligned} \quad (\text{H13})$$

with a two-level truncation in the FTT.

3. Transformation of Lindbladian

Following Ref. [36], in the dispersive regime the reduced dynamics are described by the Lindblad master equation

$$\dot{\rho}_{\text{disp}} = -i[H_{\text{disp}}, \rho_{\text{disp}}] + \gamma_{b\downarrow} \mathcal{D}[b] \rho_{\text{disp}} + \gamma_{c\downarrow} \mathcal{D}[c] \rho_{\text{disp}}.$$

Here $\gamma_{b\downarrow}$ is the FTT relaxation rate, and $\gamma_{c\downarrow} = \frac{g^2}{\Delta_{bc}^2} \gamma_{b\downarrow}$ is the Purcell-induced cavity decay rate. The cavity relaxation is assumed to be Purcell-limited in state-of-the-art hardware designs [5].

In the rotating frame defined by U_R , the equation retains the same Lindblad structure:

$$\dot{\rho}_R = -i[H_R, \rho_R] + \gamma_{b\downarrow} \mathcal{D}[b] \rho_R + \gamma_{c\downarrow} \mathcal{D}[c] \rho_R, \quad (\text{H14})$$

where $\rho_R = U_R^\dagger \rho_{\text{disp}} U_R$.

Effective dissipation in the driven frame is obtained by transforming the jump operators with U_d . The FTT operator is first truncated to the three lowest levels ($|g\rangle, |e\rangle, |f\rangle$), so that $b \approx |g\rangle\langle e| + \sqrt{2}|e\rangle\langle f|$, and the drive transformation $U_d = e^{D_d}$ is applied. Projecting the result to the two-level FTT subspace then yields

$$\begin{aligned} e^{-D_d} b e^{D_d} &\approx b + [D_d, b] + \frac{1}{2} [D_d, [D_d, b]] \\ &\approx \sigma^- + \sum_m \Omega_0 \left(-\frac{1}{\Delta_m} |m\rangle\langle m| \right. \\ &\quad \left. + \left[\frac{2}{\Delta_m} - \frac{2}{K} \right] \sigma^\dagger \sigma |m\rangle\langle m| \right) \\ &\quad + \sum_m \Omega_0^2 \left[\left(\frac{1}{K\Delta_m} - \frac{1}{\Delta_m^2} \right) \sigma^+ |m\rangle\langle m| \right. \\ &\quad \left. + \left(\frac{2}{K\Delta_m} - \frac{1}{\Delta_m^2} \right) \sigma^- |m\rangle\langle m| \right], \quad (\text{H15}) \end{aligned}$$

where we truncate FTT to two levels. Similarly, the cavity operator c transforms under the influence of the drive:

$$\begin{aligned} e^{-D_d} c e^{D_d} &\approx c + \sum_m \frac{\sqrt{m+1} \Omega_0 \chi}{\Delta_m \Delta_{m+1}} ((\sigma^+ - \sigma^-) |m\rangle\langle m+1| \\ &\quad + \sum_m \sqrt{m+1} \Omega_0^2 \left(\frac{1}{\Delta_{m+1}^2} - \frac{1}{\Delta_m^2} \right) |m\rangle\langle m+1| \\ &\quad - \sum_m \sqrt{m+1} \frac{\Omega_0^2 (\chi^2 + \Delta_m \chi)}{(\Delta_m \Delta_{m+1})^2} \sigma^+ \sigma^- |m\rangle\langle m+1| \end{aligned} \quad (\text{H16})$$

In this driven frame, the Lindblad master equation becomes

$$\begin{aligned} \dot{\rho}_d &= -i[H_d, \rho_d] + \gamma_{b\downarrow} \mathcal{D}[e^{-D_d} b e^{D_d}] \rho_d \\ &\quad + \gamma_{c\downarrow} \mathcal{D}[e^{-D_d} c e^{D_d}] \rho_d \\ &\approx -i[H_d, \rho_d] + \\ &\quad \gamma_{b\downarrow} \mathcal{D} \left[\sigma^- + \sum_m \Omega_0^2 \left(\frac{2}{K\Delta_m} - \frac{1}{\Delta_m^2} \right) \sigma^- |m\rangle\langle m| \right] \rho_d \\ &\quad + \gamma_{b\downarrow} \mathcal{D} \left[-\sum_m \frac{\Omega_0}{\Delta_m} |m\rangle\langle m| \right] \rho_d \\ &\quad + \gamma_{b\downarrow} \mathcal{D} \left[\sum_m \Omega_0^2 \left(\frac{1}{K\Delta_m} - \frac{1}{\Delta_m^2} \right) \sigma^+ |m\rangle\langle m| \right] \rho_d \\ &\quad + \sum_m \gamma_{c\downarrow, m} \mathcal{D}[|m\rangle\langle m+1|] \rho_d \\ &\quad + \mathcal{O} \left(\frac{g^2}{\Delta_{bc}^2} \frac{(m+1) \Omega_0^2 \chi^2}{(\Delta_m \Delta_{m+1})^2} \right), \quad (\text{H17}) \end{aligned}$$

where $\rho_d = U_d^\dagger \rho_R U_d$. The effective cavity decay rate $\gamma_{c\downarrow, m}$ is perturbatively modified by the drive:

$$\gamma_{c\downarrow, m} \approx (m+1) \left(1 - \Omega_0^2 \left[\frac{1}{\Delta_{m+1}^2} - \frac{1}{\Delta_m^2} \right] \right) \gamma_{c\downarrow}. \quad (\text{H18})$$

Equation (H17) uses the secular approximation, valid when the relevant detunings and mode frequencies exceed the decay rates ($\Delta_{bd}, \omega_c \gg \gamma_{b\downarrow}$). In the interaction picture, the transformed jump operator decomposes into spectral components $\sum_k L_k$ oscillating at distinct frequencies ω_k . When $\gamma \ll |\omega_j - \omega_k|$, rapidly oscillating cross-terms can be neglected, so that dissipation channels decouple:

$$\gamma \mathcal{D} \left[\sum_k L_k \right] \rho \approx \sum_k \gamma \mathcal{D}[L_k] \rho. \quad (\text{H19})$$

4. Further approximation under two regimes

We have transformed the Lindbladian in Eq. (18) into an effective drive frame, yielding the dissipators in Eq. (H17) and the noise Hamiltonian in Eq. (H13). We then derive simplified expressions for the Lindbladian and H_{noise}^d in the selective and unselective regimes discussed in Sec. II B.

a. Unselective regime Retain up to second order of $\mathcal{O}((\Delta_{bd}/\chi)^2)$, dissipators in Eq. (H17) and the noise Hamiltonian in Eq. (H13) reduce to

$$\begin{aligned} \dot{\rho} &= -i[H_d + H_{\text{noise}}^d, \rho] + \tilde{\gamma}_{b\downarrow} \mathcal{D}[\sigma^- (1 - \lambda_\downarrow c^\dagger c)] \rho \\ &\quad + 2\gamma_{c\phi} \mathcal{D}[c^\dagger c] \rho + \gamma_{b\uparrow} \mathcal{D}[\sigma^+ (1 - \lambda_\uparrow c^\dagger c)] \rho + \tilde{\gamma}_{c\downarrow} \mathcal{D}[c] \rho, \quad (\text{H20}) \end{aligned}$$

with the noise operator

$$\begin{aligned} H_{\text{noise}}^d &\approx \delta\omega_b \left[\left(\frac{g^2}{\Delta_{bc}^2} - \frac{2\Omega_0^2 \chi}{\Delta_{bd}^3} \right) c^\dagger c \right. \\ &\quad - 2 \left(\frac{\Omega_0}{\Delta_{bd}} \right)^2 \sigma^+ \sigma^- + \left(\frac{4\Omega_0^2 \chi}{\Delta_{bd}^3} - \frac{4Kg^2}{\Delta_{bc}^3} \right) \sigma^+ \sigma^- c^\dagger c \\ &\quad \left. - \frac{\Omega_0}{\Delta_{bd}} \sigma_x + \frac{\Omega_0 \chi}{\Delta_{bd}^2} \sigma_x c^\dagger c \right], \quad (\text{H21}) \end{aligned}$$

and normalized FTT relaxation rate, pure cavity dephasing rate, FTT excitation rate, normalized cavity decay rate respectively:

$$\tilde{\gamma}_{b\downarrow} \approx \gamma_{b\downarrow} \left[1 + 2 \left(\frac{2\Omega_0^2}{K\Delta_{bd}} - \frac{\Omega_0^2}{\Delta_{bd}^2} \right) \right] + \mathcal{O} \left(\frac{\Omega_0^4}{\Delta_{bd}^4} \right) \quad (\text{H22})$$

$$\gamma_{c\phi} = \gamma_{b\downarrow} \left(\frac{\Omega_0 \chi}{\Delta_{bd}^2} \right)^2 / 2 \quad (\text{H23})$$

$$\gamma_{b\uparrow} = \gamma_{b\downarrow} \left[\Omega_0^2 \left(\frac{1}{K\Delta_{bd}} - \frac{1}{\Delta_{bd}^2} \right) \right]^2 \quad (\text{H24})$$

$$\tilde{\gamma}_{c\downarrow} \approx \gamma_{c\downarrow} \left(1 + \frac{2\chi \Omega_0^2}{\Delta_{bd}^3} + \mathcal{O} \left(\frac{\chi^2}{\Delta_{bd}^2} \right) \right). \quad (\text{H25})$$

When small parameters

$$\lambda_{\downarrow} \approx \frac{2\chi}{\Delta_{bd}} \Omega_0^2 \left(\frac{1}{K\Delta_{bd}} - \frac{1}{\Delta_{bd}^2} \right) + \mathcal{O}\left(\frac{\Omega_0^4}{\Delta_{bd}^4}\right) \quad (\text{H26})$$

$$\lambda_{\uparrow} = \frac{\chi}{\Delta_{bd}} \left(\frac{2K - \Delta_{bd}}{K - \Delta_{bd}} \right). \quad (\text{H27})$$

The cavity-photon dependence of the decoherence rates originates from the drive-induced nonlinearity. For higher cavity occupations, the AC-Stark-shifted detuning Δ_m increases, which suppresses the FTT heating rate and drives the relaxation rate toward $\tilde{\gamma}_{b\downarrow}$. In the main text, we neglect this photon-number dependence to keep the discussion focused on the leading scaling. Using Eq. (14), we then recover the simplified master equation quoted in subsection III A.

b. Selective regime. In the selective regime, the condition $|\Delta_{m^*}/\chi| \ll 1$ holds. Combining the relation $\chi \approx 2\frac{g^2}{\Delta_{bc}^2}K$ with the hierarchy of energy scales gives

$$|\Omega_0| \ll |\Delta_{m^*}| \ll |\chi| \ll |K|. \quad (\text{H28})$$

Accordingly, drive-induced corrections to the dissipators are retained up to $\mathcal{O}(\Omega_0/K)$. Terms of the form $\Omega_0/(\Delta_{m^*} + m\chi)$ in Eq. (H17) scale as $\mathcal{O}(\Omega_0/\chi)$ for $m \neq m^*$ (since $|\Delta_{m^*} + m\chi| \sim |m\chi|$), whereas the resonant contribution scales as Ω_0/Δ_{m^*} for $m = m^*$. To leading order of Ω_0/Δ_{m^*} , the master equation reduces to

$$\begin{aligned} \dot{\rho} \approx & -i[H_d + H_{\text{noise}}^d, \rho] \\ & + \gamma_{b\downarrow} \mathcal{D}\left[\sigma^- - \frac{\Omega_0^2}{\Delta_{m^*}^2} \sigma^- |m^*\rangle\langle m^*|\right] \rho + 2\gamma_{c\downarrow}^{\text{sel}} \mathcal{D}[|m^*\rangle\langle m^*|] \rho \\ & + \gamma_{b\uparrow}^{\text{sel}} \mathcal{D}[\sigma^+ |m^*\rangle\langle m^*|] \rho + \sum_m \tilde{\gamma}_{c\downarrow, m}^{\text{sel}} \mathcal{D}[|m\rangle\langle m+1|] \rho, \end{aligned} \quad (\text{H29})$$

and the noise Hamiltonian is

$$\begin{aligned} H_{\text{noise}}^d \approx & \delta\omega_b(t) \left[\frac{g^2}{\Delta_{bc}^2} c^\dagger c + \frac{g^2}{\Delta_{bc}^2} |m^*\rangle\langle m^*| \right. \\ & + \sigma^+ \sigma^- \left(\frac{4Kg^2}{\Delta_{bc}^3} c^\dagger c + 2\frac{g^2}{\Delta_{bc}^2} |m^*\rangle\langle m^*| \right) \\ & \left. - \frac{g}{\Delta_{bc}} \sigma^+ |m^*\rangle\langle m^*| + \text{h.c.} \right]. \end{aligned} \quad (\text{H30})$$

Here, the selective-regime sweet-spot condition Eq. (16) has been used. At this operating point, the longitudinal component of H_{noise}^d produces no dephasing between $|g, m^*\rangle$ and $|g, m^* + 1\rangle$. Meanwhile residual cavity dephasing from AC-Stark-shift fluctuations induced by $\delta\omega_b(t)$ is negligible for the dynamics of interest because the FTT remains near its ground state when initialized in $|g\rangle$. The transverse component instead induces depolarization; within a Markovian approximation, the corresponding excitation rate due to $1/f$ noise is $\gamma_{b\uparrow}^{1/f, \text{sel}} = (\partial\omega_b/\partial\Phi)^2 (g/\Delta_{bc})^2 S_{1/f}(\Delta_{bd})$.

The dissipative processes in the effective drive frame can be summarized as follows: (i) FTT relaxation with a cavity-state-dependent correction, together with cavity decay at a renormalized rate $\gamma_{c\downarrow, m}^{\text{sel}} = \gamma_{c\downarrow}^{\text{sel}} (1 + \delta_{m^*, m} g^2/\Delta_{bc}^2)$, where $\gamma_{c\downarrow}^{\text{sel}} = (g^2/\Delta_{bc}^2)\gamma_{b\downarrow}$ is the Purcell decay rate; (ii) FTT heating at rate $\gamma_{b\uparrow}^{\text{sel}} = (g^4/\Delta_{bc}^4)\gamma_{b\downarrow}$ when the cavity occupies $|m^*\rangle$; and (iii) pure dephasing between $|m^*\rangle$ and $|m^* + 1\rangle$ with rate $\gamma_{c\phi}^{\text{sel}} = (g^2/\Delta_{bc}^2)\gamma_{b\downarrow}^{\text{sel}}/2$.

Dephasing between $|g, m^*\rangle$ and $|g, m^* + 1\rangle$ is therefore limited by cavity pure dephasing at rate $\gamma_{c\phi}^{\text{sel}} \propto g^2/\Delta_{bc}^2$, and by FTT heating at rate $\gamma_{b\uparrow}^{\text{sel}} \propto 1/\Delta_{bd}$ due to photon shot noise in the selective regime.

Appendix I: Numerical details of obtaining dephasing rates

This section details the numerical procedure used to extract the dephasing rates in Sec. III C. The dynamics are simulated with the full Hamiltonian $H_{\text{full}}(\delta\Phi)$ represented in a truncated oscillator (Fock) basis and evolved under the Lindblad master equation

$$\dot{\rho} = -i[H_{\text{full}}(\delta\Phi(t)), \rho] + \gamma_{b\downarrow} \mathcal{D}[b]\rho, \quad (\text{I1})$$

where $\mathcal{D}[b]\rho = b\rho b^\dagger - \frac{1}{2}\{b^\dagger b, \rho\}$.

a. Generation of $1/f$ noise. Flux noise is incorporated through stochastic trajectories $\delta\Phi(t)$ whose spectrum follows a $1/f$ power spectral density. For each trajectory, Eq. (I1) is integrated to obtain $\rho(t)$, and all reported curves are computed from ensemble averages of the corresponding observables. The noise realizations $\delta\Phi(t)$ are generated using the frequency-domain synthesis method of Ref. [43]: Fourier components are assigned random phases and Gaussian-distributed amplitudes with variances proportional to the target $1/f$ spectrum, followed by an inverse Fourier transform to obtain $\delta\Phi(t)$ in the time domain. Fig. 7 shows the resulting power spectral densities for 100 representative trajectories. This ensemble size is sufficient for the quantities of interest to converge, as indicated by the small error bars in Fig. 3(a) and Fig. 8(b).

The total simulated evolution time is $T \sim 10 \mu\text{s}$, and the infrared cutoff is set to $\omega_{\text{IR}}/2\pi = 10 \text{ kHz}$ (equivalently $f_{\text{min}} = 10^{-5} \text{ GHz}$). This choice satisfies $f_{\text{min}} \ll 1/T \approx 100 \text{ kHz}$, capturing the relevant slow-noise dynamics within the simulated window while avoiding the infrared divergence of an ideal $1/f$ spectrum. The ultraviolet cutoff is set to $f_{\text{max}} = 1 \text{ GHz}$; the extracted dephasing rates are insensitive to further increases of f_{max} .

b. Extraction of dephasing time. The states with enhanced dephasing discussed in main text are proxies for the corresponding Floquet states, as shown in Appendix C. In the numerics, Floquet modes are labeled by their correspondence to bare states as described in Appendix D. The two Floquet states used for the cavity

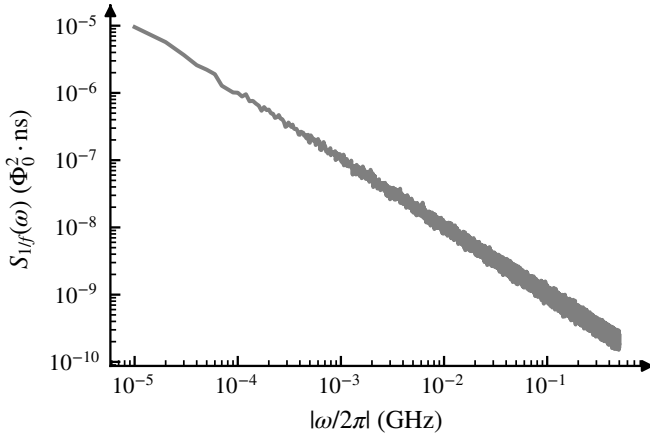


Figure 7. Power spectral density of $\delta\Phi(t)$ for 100 representative $1/f$ noise trajectories generated with the cutoffs specified below.

dephasing analysis in Sec. III C are denoted $|\phi_{g,0}(t)\rangle$ and $|\phi_{g,1}(t)\rangle$.

Relaxation and dephasing are extracted from dynamics evolving from two initial states. Relaxation is characterized by preparing

$$\rho(0) = |\phi_{g,1}(0)\rangle\langle\phi_{g,1}(0)|$$

and monitoring the return toward $|\phi_{g,0}(t)\rangle$. Dephasing is characterized by preparing the equal superposition

$$\rho(0) = |+\rangle\langle+|, \quad |+\rangle = \frac{1}{\sqrt{2}}(|\phi_{g,0}(0)\rangle + |\phi_{g,1}(0)\rangle),$$

and tracking the decay of the corresponding coherence. The relevant quantities are

$$\text{Tr}[\rho(t) P_{g,0}(t)] \quad \text{and} \quad \text{Tr}[\rho(t) X(t)],$$

where

$$P_{g,0}(t) = |\phi_{g,0}(t)\rangle\langle\phi_{g,0}(t)|, X(t) = |\phi_{g,1}(t)\rangle\langle\phi_{g,0}(t)| + \text{H.c.}$$

Relaxation and dephasing times are obtained by fitting ensemble-averaged signals. Specifically, T_1 is extracted from the rise of the ground-state population by fitting $\text{Tr}[P_{g,0}(t) \bar{\rho}(t)]$ to $1 - e^{-t/T_1}$, where $\bar{\rho}(t)$ denotes the ensemble-averaged density matrix over different flux noise trajectories. The coherence time T_2 is extracted from the decay of the superposition signal by fitting $\text{Tr}[P_+(t) \bar{\rho}(t)]$ to e^{-t/T_2} (with the appropriate normalization convention for the plotted data). The pure dephasing time then follows from

$$\frac{1}{T_\phi} = \frac{1}{T_2} - \frac{1}{2T_1}, \quad (12)$$

and we report the total dephasing rate as $\Gamma_{\phi,\text{total}} = 1/T_\phi$. The data in Fig. 3 use the same system parameters and level truncations as Appendix D. Unless otherwise stated, the fitting windows are $1 \mu\text{s}$ for T_1 and $5 \mu\text{s}$ for T_2 ; extending these windows does not change the extracted rates.

Appendix J: Rates in the unselective regime

1. Dephasing rates

This subsection benchmarks the analytical dephasing rates in Eq. (24) against numerical simulations. The numerical reference rate for $\gamma_{b\uparrow}^{\text{un}} + \gamma_{c\phi}^{\text{un}}$ is extracted using the procedure in App. I, with the flux-noise drive set to $\delta\Phi(t) = 0$.

In the large-detuning regime, the reference rate is well captured by the analytical approximation in Fig. 8(a):

$$\gamma_{b\uparrow}^{\text{un}} + \gamma_{c\phi}^{\text{un}} \quad (J1)$$

$$= \frac{\gamma_{b\downarrow}}{16} \left(\frac{\Delta_{bd}}{K} \right)^2 \left(1 - \frac{2\Delta_{bd}}{K} \right) + \frac{\gamma_{b\downarrow}}{2} \left(\frac{g}{\Delta_{bc}} \right)^4 \frac{K}{\Delta_{bd}}. \quad (J2)$$

In this limit, dephasing is dominated by photon-shot noise from heating, i.e., $\gamma_{b\uparrow}^{\text{un}}$, while $\gamma_{c\phi}^{\text{un}}$ is parametrically suppressed at large detuning, consistent with Eq. (J2).

As the detuning decreases, the numerical and analytical results deviate because Eq. (J2) assumes $|\Delta_{bd}/\chi| \gg 1$ (here $\chi/2\pi \simeq -2 \text{ MHz}$). In this regime, the total dephasing rate increases and $\gamma_{c\phi}^{\text{un}}$ becomes the dominant contribution.

Finally, the flux-noise-induced depolarization rate is benchmarked numerically against

$$\gamma_{b\uparrow}^{1/f,\text{un}} = \left(\frac{\partial\omega_b}{\partial\Phi} \right)^2 \frac{S_0^2}{4K}. \quad (J3)$$

The simulation is initialized in the Floquet ground state $|\phi_{g0}(0)\rangle$, and the excitation rate is extracted by fitting the average overlap $\text{Tr}[\rho |\phi_{g0}(0)\rangle\langle\phi_{g0}(0)|]$ to $\frac{1}{2}(1 - e^{-2\gamma_{b\uparrow}^{1/f,\text{un}} t})$. Sweeping the noise amplitude S_0 yields a quadratic scaling, and the numerical results follow the analytical prediction (black dashed line) closely [Fig. 8(b)]. The small error bars in the figure indicate statistical convergence for the chosen ensemble size ($N = 100$ trajectories).

2. Decay rates

This subsection shows that the cavity decay rate is only weakly affected by the drive. For the parameters used to generate Fig. 3, the undriven decay rate is $4.75 \times 10^{-7} \text{ ns}^{-1}$. Under drive, the decay rate remains nearly unchanged, as shown in Fig. 8(c).

Appendix K: Adiabatic transition

In the adiabatic frame, we introduce a time-dependent unitary transformation $U(\Omega_t)$ that diagonalizes the instantaneous Hamiltonian $H(\Omega_t)$. The corresponding ef-

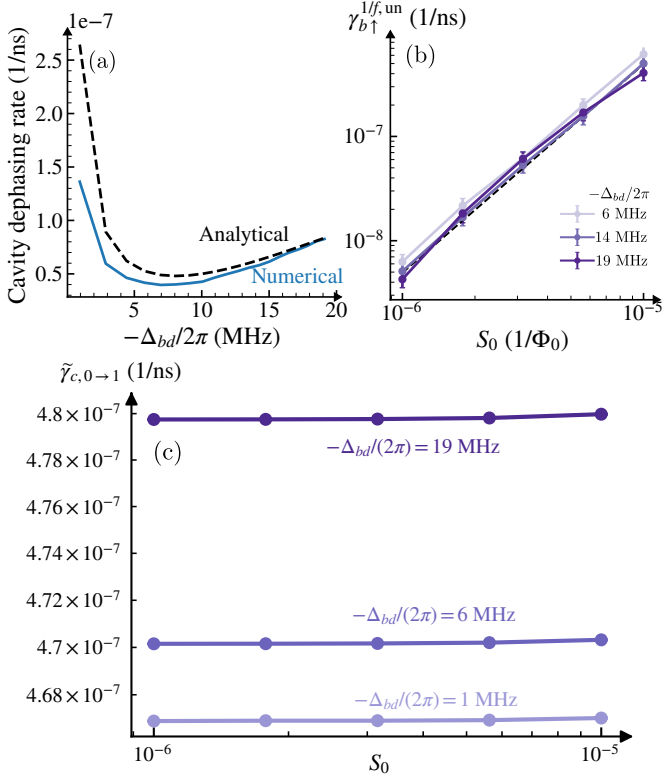


Figure 8. Numerical validation of rates in the unselective regime. (a) Reference dephasing rate $\gamma_{b\uparrow}^{un} + \gamma_{c\phi}^{un}$ extracted from the procedure in App. I with $\delta\Phi(t) = 0$, compared with the analytical approximation in Eq. (J2). (b) Flux-noise-induced depolarization rate $\gamma_{b\uparrow}^{1/f,un}$ versus noise amplitude S_0 . The quadratic scaling in S_0 and agreement with $\gamma_{b\uparrow}^{1/f,un} = \left(\frac{\partial\omega_b}{\partial\Phi}\right)^2 \frac{S_0^2}{4K}$ (black dashed line) are observed. (c) Cavity decay rate with sweet-spot drive.

fective Hamiltonian is

$$H_{\text{eff}}(\Omega_t) = U^\dagger(\Omega_t) \left(H(\Omega_t) - i\hbar \frac{\partial}{\partial t} \right) U(\Omega_t) \quad (\text{K1})$$

$$= H_d(\Omega_t) + V(\Omega_t).$$

The diagonal part H_d contains the instantaneous energy eigenvalues $E_j(\Omega_t)$:

$$H_d(\Omega_t) = \sum_j E_j(\Omega_t) |j\rangle\langle j|. \quad (\text{K2})$$

The coupling term V originates from the induced gauge contribution $i\hbar U^\dagger \dot{U}$ and captures the effect of the time-dependent driving. Splitting V into off-diagonal and di-

agonal components gives

$$V = \underbrace{-i\hbar \sum_{j \neq j'} \frac{\langle j' | U^\dagger \dot{U} U | j \rangle}{E_j - E_{j'}} |j'\rangle\langle j|}_{V_{od}} + \underbrace{-i\hbar \sum_j \langle j | U^\dagger \dot{U} | j \rangle |j\rangle\langle j|}_{V_d}. \quad (\text{K3})$$

where the diagonal terms contribute to the Berry phase, while the off-diagonal terms drive transitions.

In our case of interest, the goal is to adiabatically evolve into the desired Floquet states. When counter-rotating terms are negligible, this evolution is equivalent to adiabatic dynamics governed by the rotating-frame Hamiltonian

$$H_R(t) = \Delta_{bd} b^\dagger b + \frac{K}{2} b^{\dagger 2} b^2 + \bar{\omega}_c c^\dagger c + \chi b^\dagger b c^\dagger c + \Omega_I(t) (b + b^\dagger). \quad (\text{K4})$$

Under the drive $\Omega_I(t)$, the Hamiltonian in the instantaneous eigenbasis can be written as $H_{\text{eff}} = H_d(\Omega_I) + V(\Omega_I)$. For the driven operating regime of interest, the diagonal contribution is well approximated by Eq. (11)

$$H_d(\Omega_I) \approx \left(\Delta_{bd} - \frac{\Omega_I^2}{\kappa} \right) \sigma^+ \sigma^- + \left(\bar{\omega}_c + \frac{\Omega_I^2 \chi}{\Delta_{bd}^2} \right) c^\dagger c + \chi \left(1 - 2 \frac{\Omega_I^2}{\Delta_{bd}^2} \right) \sigma^+ \sigma^- c^\dagger c. \quad (\text{K5})$$

Nonadiabatic leakage is generated by the time dependence of the drive through the off-diagonal term V_{od} . In particular,

$$U^\dagger \dot{H}_R U = \dot{\Omega}_I U^\dagger (b^\dagger + b) U \stackrel{(\text{H15})}{\approx} \dot{\Omega}_I \sigma_x + \mathcal{O}\left(\frac{\dot{\Omega}_I}{\Delta_{bd}}\right). \quad (\text{K6})$$

which yields the effective coupling between $|g\rangle$ and $|e\rangle$,

$$V_{od}(\Omega_I) \approx \frac{\dot{\Omega}_I}{\Delta_{bd}} \sigma_y. \quad (\text{K7})$$

Suppressing leakage requires the nonadiabatic coupling to remain small compared to the instantaneous gap,

$$\frac{\dot{\Omega}_I}{\Delta_{bd}} \ll \Delta_{bd}. \quad (\text{K8})$$

For a ramp of characteristic amplitude Ω_0 executed over a ramp time τ (so that $\dot{\Omega}_I \sim \Omega_0/\tau$), this implies the lower bound

$$\tau \gg \frac{\Omega_0}{\Delta_{bd}^2} \stackrel{(14)}{=} \sqrt{\frac{1}{4K\Delta_{bd}}}. \quad (\text{K9})$$

Increasing Δ_{bd} can reduce τ to obtain the desired state.

To bypass this speed limit, we implement a shortcut to adiabaticity using an active quadrature control field Ω_Q which is DRAG-style. Specifically, we augment the drive Hamiltonian by

$$H_{\text{total}} = H_R - i\Omega_Q(b - b^\dagger), \quad (\text{K10})$$

and choose the quadrature amplitude to cancel the non-adiabatic coupling,

$$\Omega_Q = -\frac{\dot{\Omega}_I}{\Delta_{bd}}. \quad (\text{K11})$$

The effective Hamiltonian in the adiabatic frame then becomes

$$\begin{aligned} U^\dagger \left(H_R - i\Omega_Q(b - b^\dagger) - i\frac{\partial}{\partial t} \right) U \\ \approx H_d + V_d + \frac{\dot{\Omega}_I}{\Delta_{bd}}\sigma_y - \frac{\dot{\Omega}_I}{\Delta_{bd}}\sigma_y \\ = H_d + V_d, \end{aligned} \quad (\text{K12})$$

so the evolution remains diagonal in the instantaneous eigenbasis even for fast ramps.

Numerical simulations confirm the qualitative analysis above. We compute the state-preparation infidelity at the end of the ramp as a function of the ramp time τ for several detunings Δ_{bd} . The infidelity is defined as the average loss of overlap with the target Floquet states,

$$\mathcal{I}(\tau) = 1 - \frac{1}{2} [\text{Tr}(P_{g0}(\tau) \bar{\mathcal{E}}_\tau(\rho_{0,g0})) + \text{Tr}(P_{g1}(\tau) \bar{\mathcal{E}}_\tau(\rho_{0,g1}))]. \quad (\text{K13})$$

where $P_{g0}(\tau) = |\phi_{g0}(\tau)\rangle\langle\phi_{g0}(\tau)|$ and $P_{g1}(\tau) = |\phi_{g1}(\tau)\rangle\langle\phi_{g1}(\tau)|$ project onto the desired Floquet states at the end of the pulse. The map $\bar{\mathcal{E}}_\tau(\cdot) = \langle \mathcal{E}_\tau^{\delta\Phi}(\cdot) \rangle_{\delta\Phi}$ denotes the evolution averaged over flux trajectories $\delta\Phi(t)$, with each trajectory generated by the master equation Eq. (I1). The initial states are $\rho_{0,g0} = |\bar{g}0\rangle\langle\bar{g}0|$ and $\rho_{0,g1} = |\bar{g}1\rangle\langle\bar{g}1|$, where the overbar denotes eigenstates of the static system Hamiltonian in the dispersive regime.

Results are shown in Fig. 9. For a fixed detuning, the infidelity initially decreases with increasing τ because a slower ramp improves adiabaticity and suppresses coherent leakage, but it eventually increases as the longer evolution accumulates more decoherence error. For a fixed ramp time (e.g., $\tau = 50$ ns), larger detuning yields lower infidelity, due to larger instantaneous gap. Finally, for the same detuning and ramp time, DRAG outperforms a Gaussian ramp because the added Q -quadrature control enhances adiabaticity by suppressing unwanted transitions.

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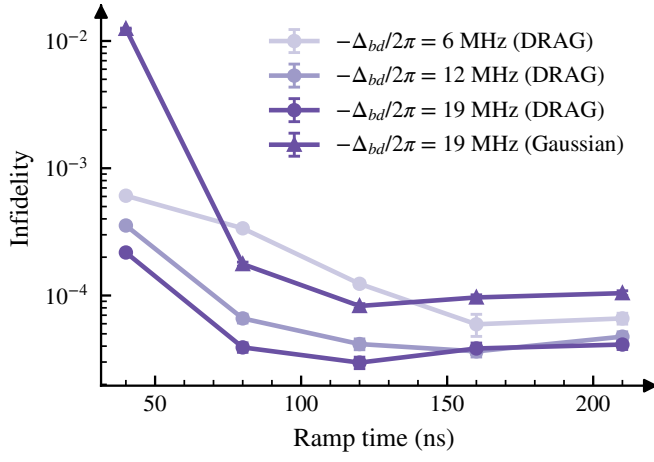


Figure 9. State-preparation infidelity $\mathcal{I}(\tau)$ versus ramp time τ for several detunings Δ_{bd} , comparing Gaussian and DRAG ramps. The non-monotonic dependence reflects the trade-off between coherent leakage from unwanted transitions (suppressed for longer τ) and decoherence (which accumulates for longer τ).